

CO2xN: Porewater Methane

Month Year Samples

2026-06-09

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##Run Information

```
cat("Run Information: Melanie Giessner ") #lets you know what section you're in
```

Run Information: Melanie Giessner

```
#set the run date & user name
sample_year <- "2026"
sample_month <- "May"
user <- "Melanie Giessner"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

#change dates
folder_path_rawdata <- paste0("Raw Data")
GHG_files <- list.files(path = folder_path_rawdata,pattern = "20260513" ,full.names = TRUE)
head(GHG_files)
```

```
## [1] "Raw Data/20260513_GENX_GHGdata.csv"
```

```
#GHG_files <- c("Raw Data/20260417_GENX_GHGdata.csv","Raw Data/20260501_GENX_GHGdata.csv" )
```

```
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/ShimadzuGC_Slope_QAQC.csv"
```

```
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_GENX_Porewater_CH4_20260513.csv"
```

#Gas in Water Calculations

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

#record any notes about the run or anything other info here:

```
run_notes <- ""
```

#Metadata File Path

```
Raw_Metadata = here("..", "GCReW", "GCReW_Project_Treatment_Metadata.csv")
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

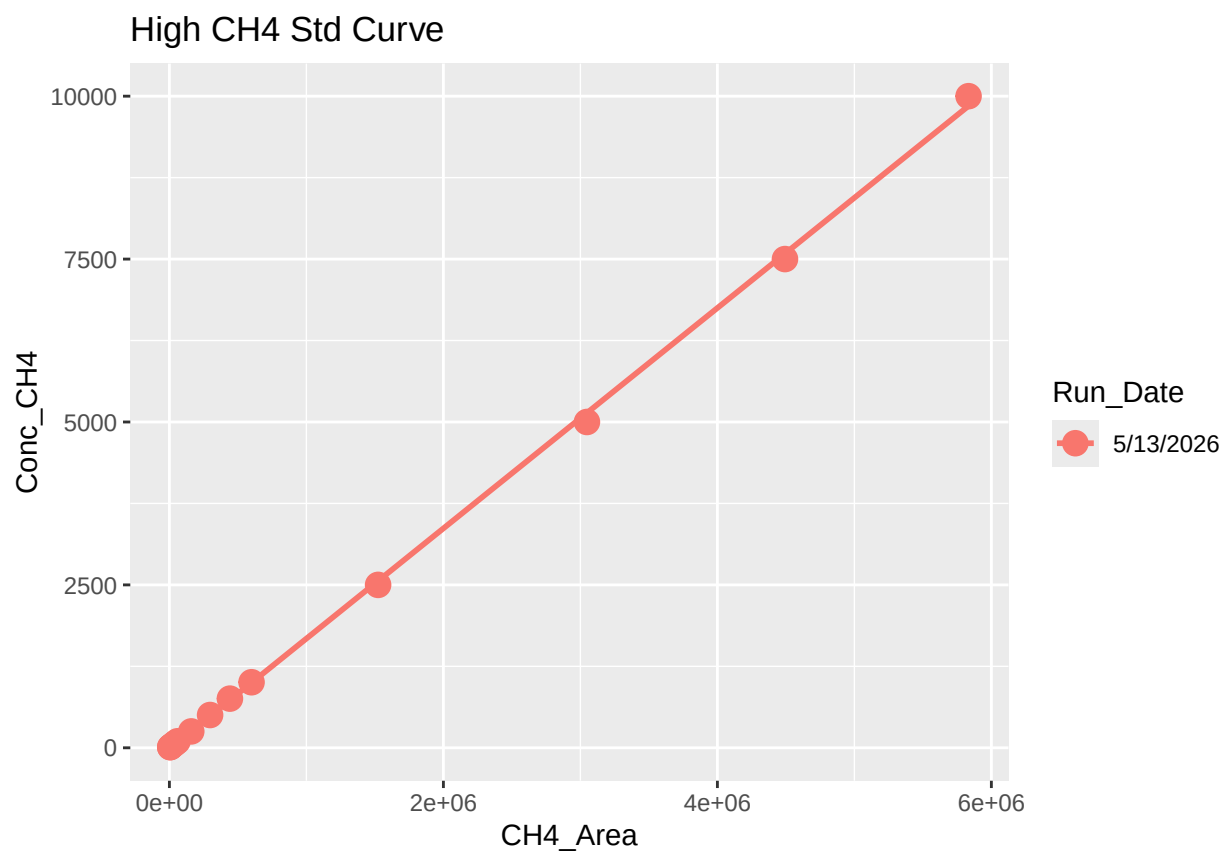
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

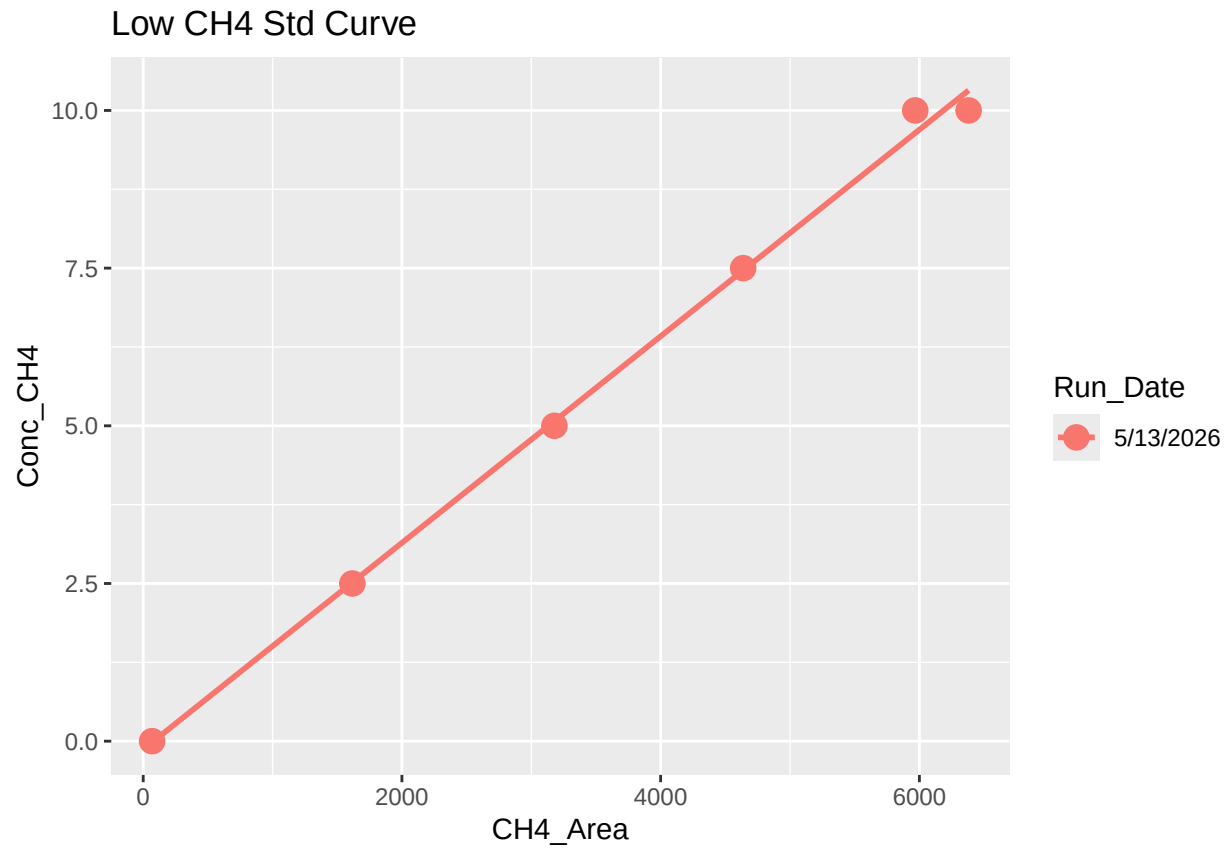
1 Read in data files

2 Assess standard curves

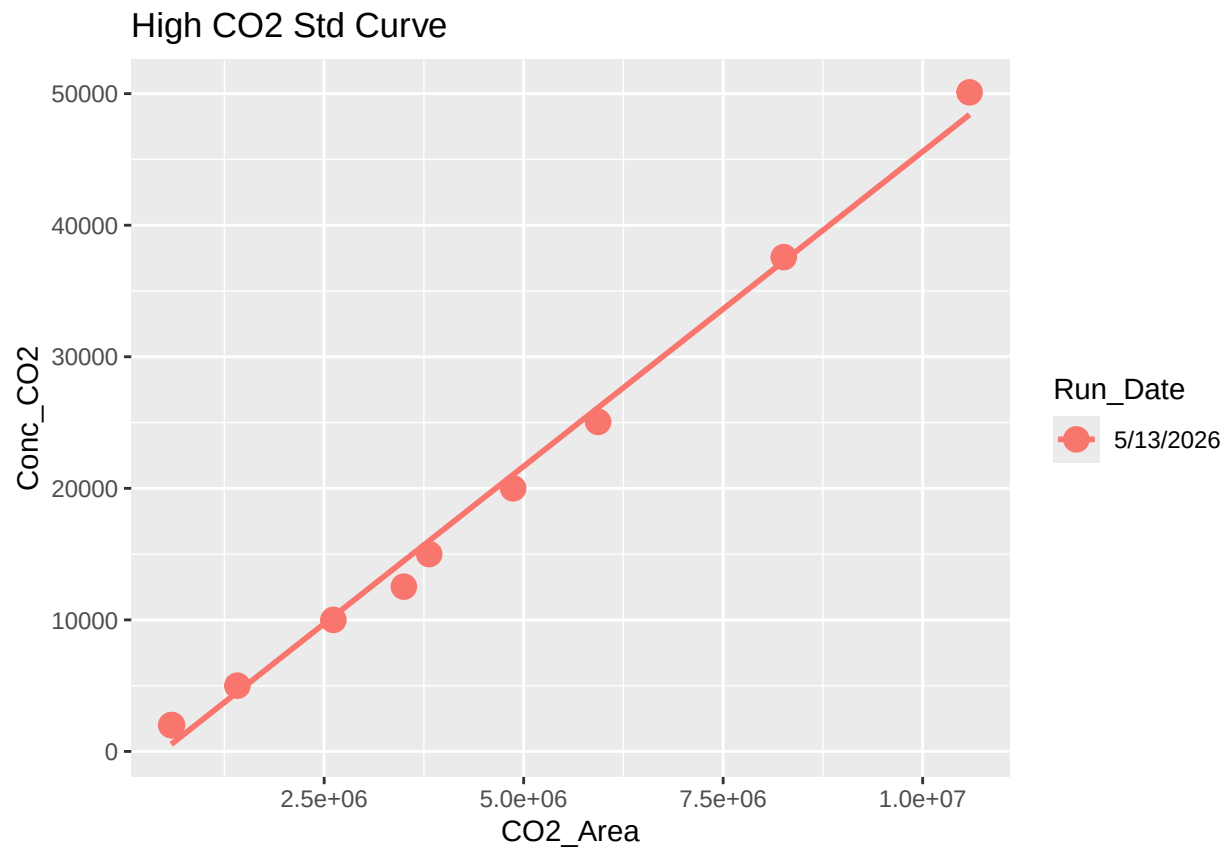
```
## 'geom_smooth()' using formula = 'y ~ x'
```



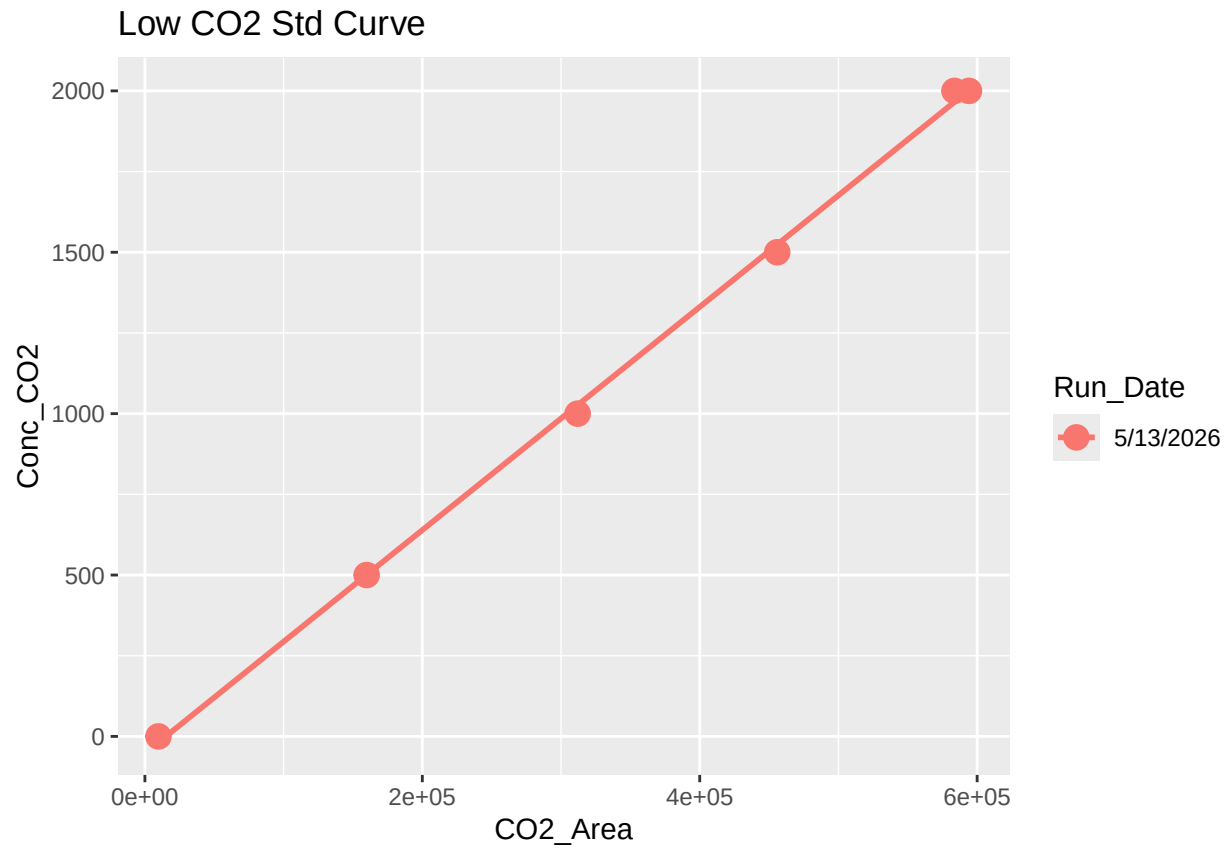
```
## 'geom_smooth()' using formula = 'y ~ x'
```



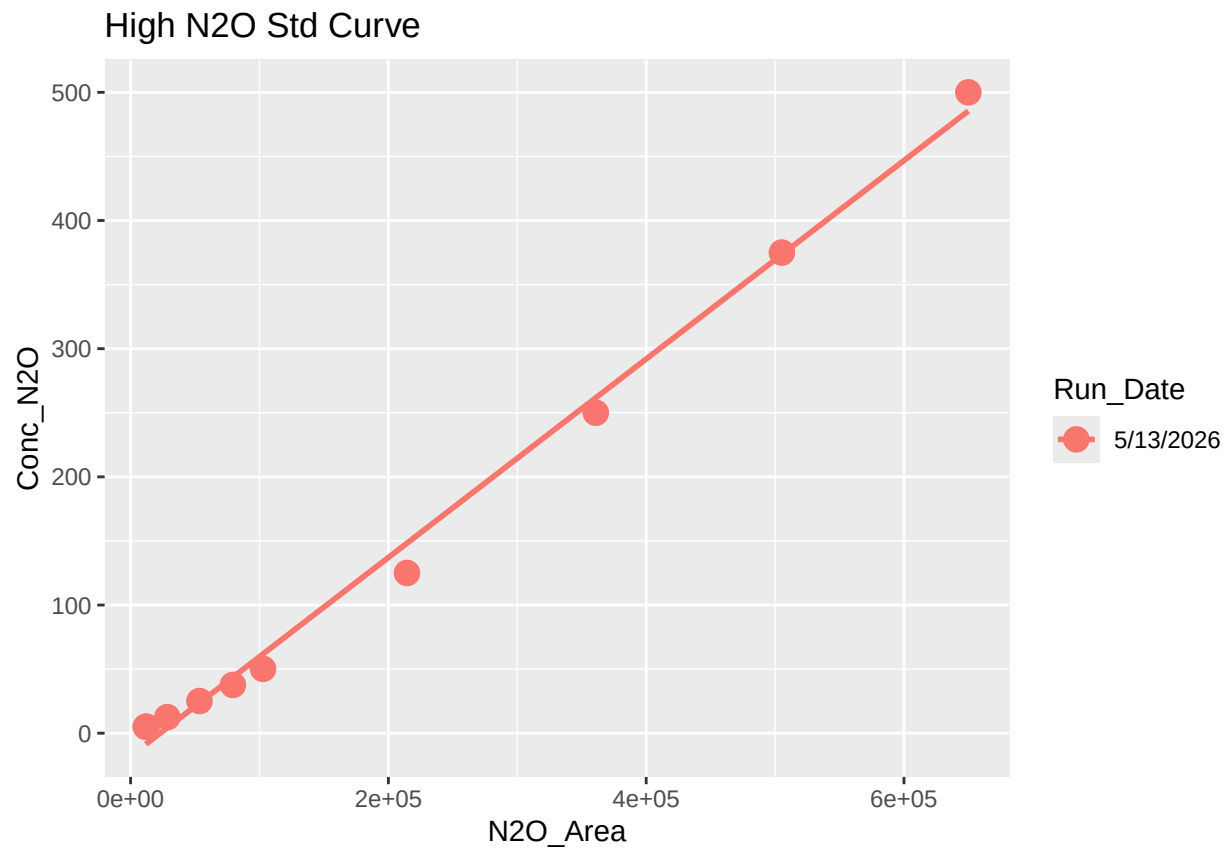
```
## Joining with 'by = join_by(Run_Date)'  
  
## [1] "High_R2 is Good-5/13/2026"  
  
## [1] "Low_R2 is Good-5/13/2026"  
  
## 'geom_smooth()' using formula = 'y ~ x'
```



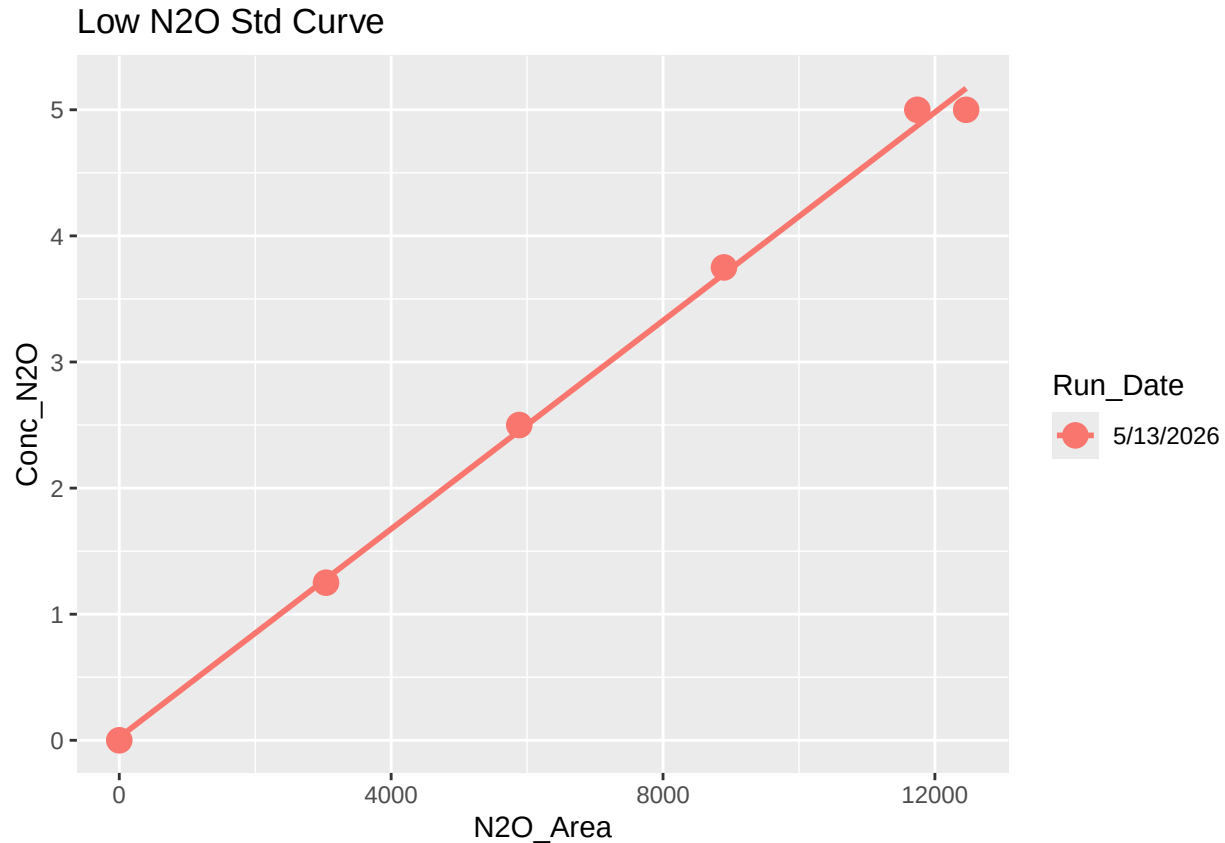
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'  
  
## [1] "High_R2 is Good-5/13/2026"  
  
## [1] "Low_R2 is Good-5/13/2026"  
  
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

```
## [1] "High_R2 is Good-5/13/2026"
```

```
## [1] "Low_R2 is Good-5/13/2026"
```

```
#add new standard curve data to log
```

```
Slope_CH4_1 <- Slope_CH4 %>% mutate(curve = paste0("Slope_CH4_", CH4_Curve), R2 = CH4_R2, slope = CH4_S
```

```
  CH4_Curve == "Low" ~ paste(CH4_low_std_crv_cutoff,"ppm"),
```

```
  CH4_Curve == "High" ~ paste(CH4_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(CH4_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_CO2_1 <- Slope_CO2 %>% mutate(curve = paste0("Slope_CO2_", CO2_Curve), R2 = CO2_R2, slope = CO2_S
```

```
  CO2_Curve == "Low" ~ paste(CO2_low_std_crv_cutoff,"ppm"),
```

```
  CO2_Curve == "High" ~ paste(CO2_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(CO2_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope_N2O_1 <- Slope_N2O %>% mutate(curve = paste0("Slope_N2O_", N2O_Curve), R2 = N2O_R2, slope = N2O_S
```

```
  N2O_Curve == "Low" ~ paste(N2O_low_std_crv_cutoff,"ppm"),
```

```
  N2O_Curve == "High" ~ paste(N2O_high_std_crv_cutoff,"ppm"),
```

```
  .default = as.character(N2O_mid_std_crv_cutoff)
```

```
) %>% dplyr::select(Analysis, Run_Date, Run_by, Curve_top, curve, R2, slope, intercept)
```

```
Slope <- rbind(Slope_CH4_1, Slope_CO2_1, Slope_N2O_1)
```

```
log <- read.csv(log_path)
log <- log[, -c(1)]
head(log)
```

```
##   Analysis Run_Date      Run_by Curve_top      curve      slope
## 1    CH4      <NA> Alia Al-Haj    10 ppm Slope_Ch4_low 0.001594508
## 2    CH4      <NA> Alia Al-Haj   100 ppm Slope_Ch4_high 0.001562326
## 3   N2O      <NA> Alia Al-Haj    5 ppm Slope_N2O_low 0.000192023
## 4   N2O      <NA> Alia Al-Haj  50.2 ppm Slope_N2O_high 0.000282245
## 5    CO2      <NA> Alia Al-Haj  2000 ppm Slope_CO2_low 0.000282245
## 6    CO2      <NA> Alia Al-Haj 20000 ppm Slope_CO2_high 0.001549502
##      intercept      R2
## 1 -0.20901179 0.9960215
## 2  0.09646243 0.9996531
## 3  0.07848462 0.9940785
## 4 -0.49925490 0.9960644
## 5 -0.49925490      NA
## 6 38.64567509 0.9996907
```

```
log <- rbind(log, Slope)
```

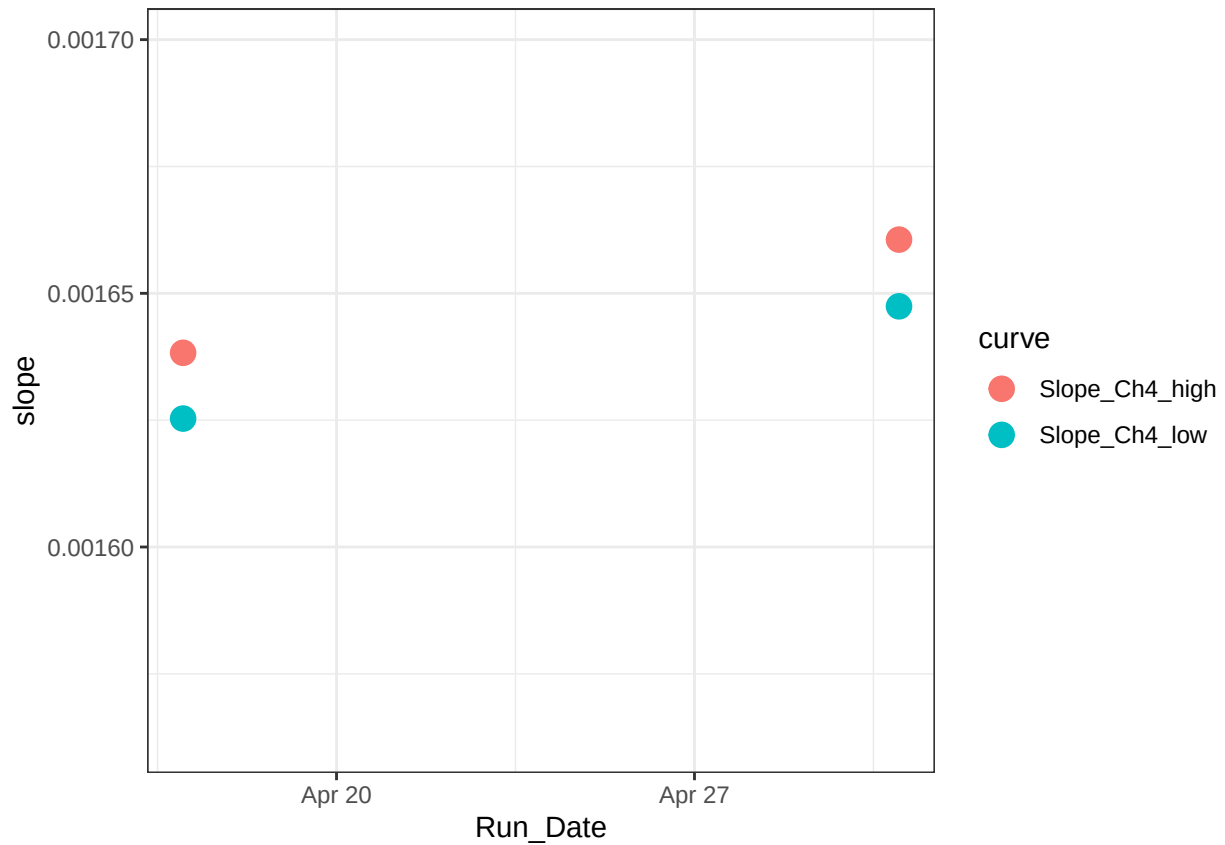
#remove possible duplicates from multiple QAQC attempts

```
unique_log <- log %>%
  mutate(Run_Date = as.Date(Run_Date, tryFormats = c("%Y-%m-%d", "%m/%d/%Y", "%m/%d/%y")), curve = gs)

Slopes_chk_CH4 <- ggplot(unique_log %>% filter(grepl("CH4", curve, ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw())
```

```
Slopes_chk_CH4
```

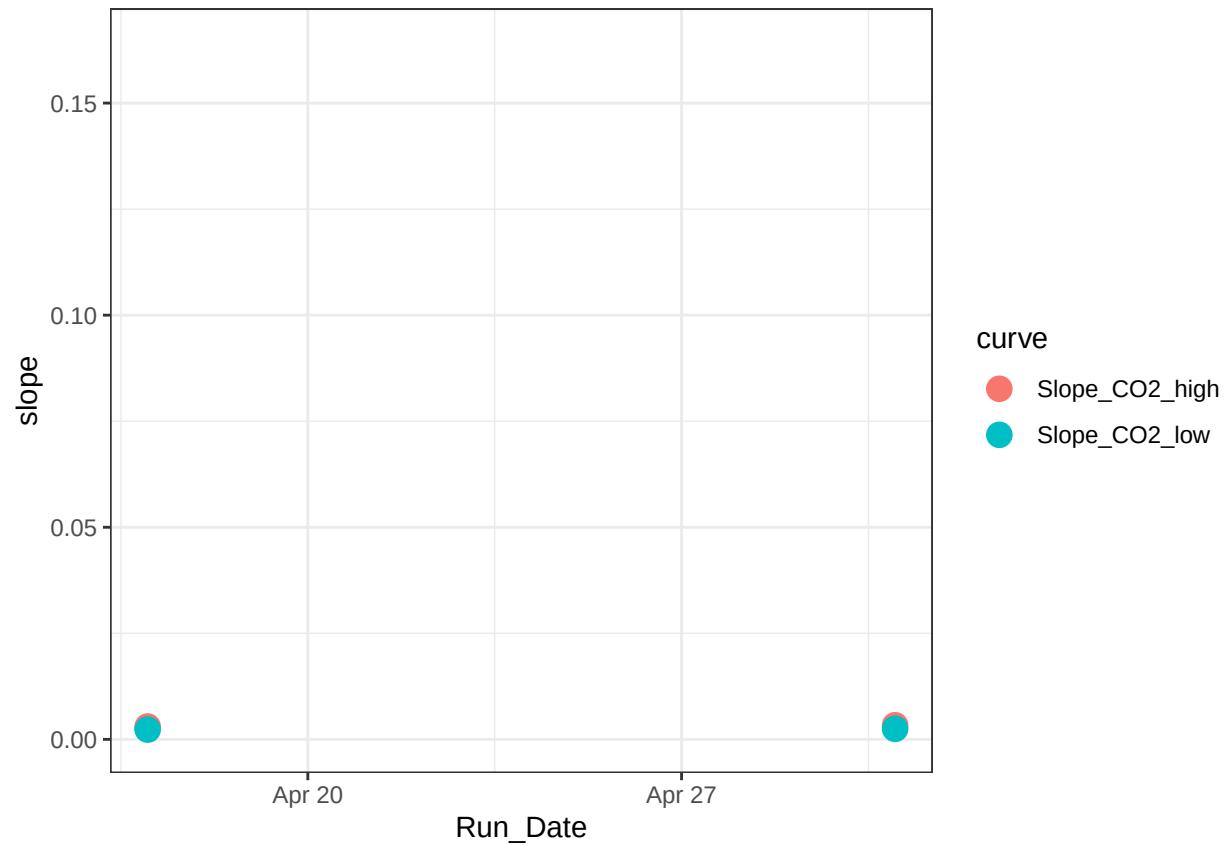
```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
Slopes_chk_CO2 <- ggplot(unique_log %>% filter( grepl("CO2",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw())
```

Slopes_chk_CO2

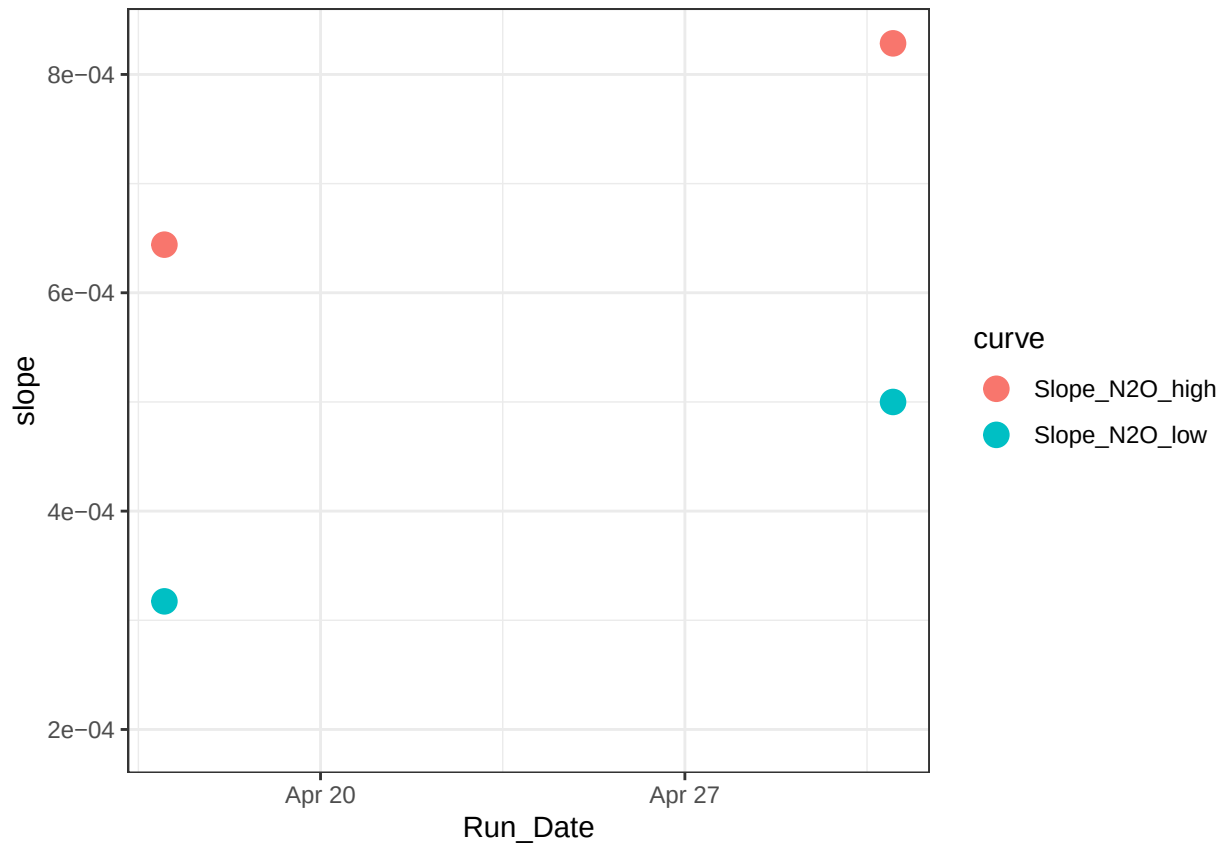
```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
Slopes_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date,
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

Slopes_chk_N20

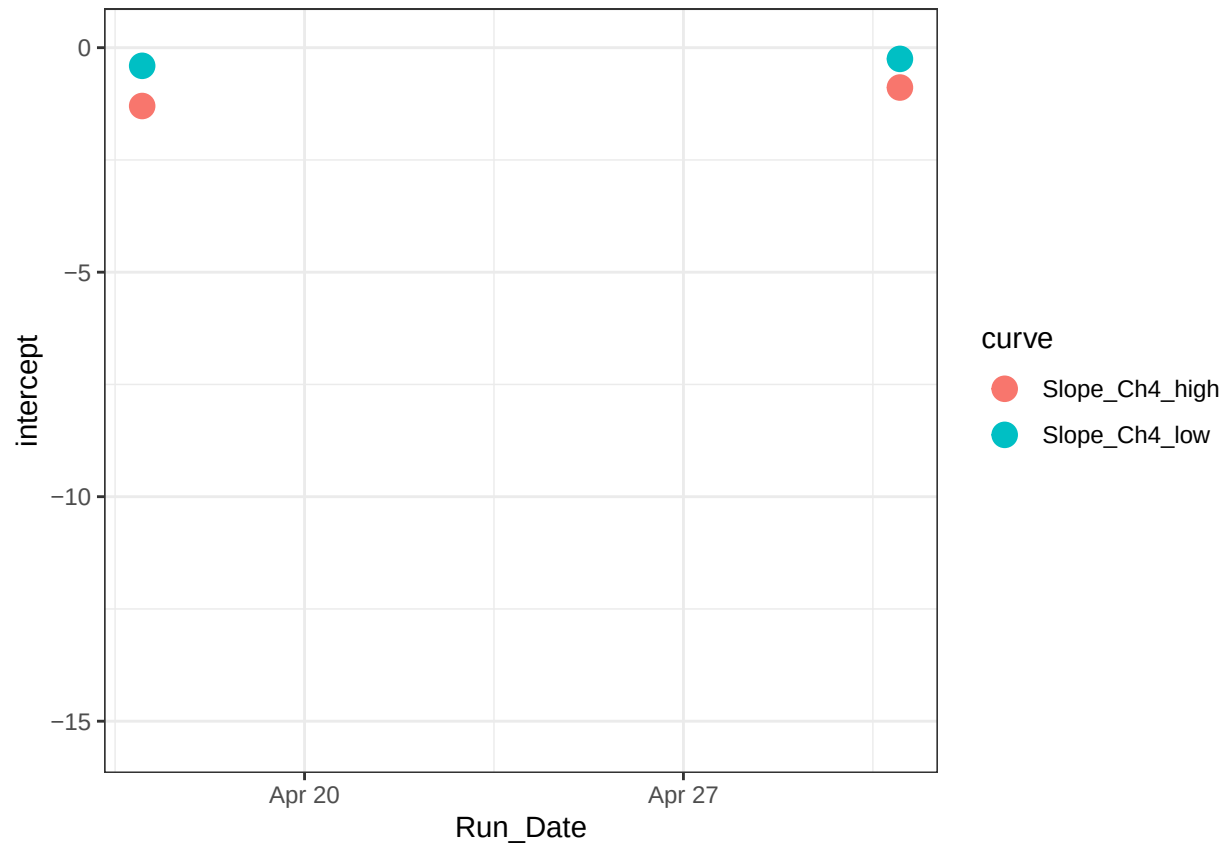
```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_CH4 <- ggplot(unique_log %>% filter( grepl("CH4",curve,ignore.case = T)), aes(x = Run_Date, y = slope)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_CH4
```

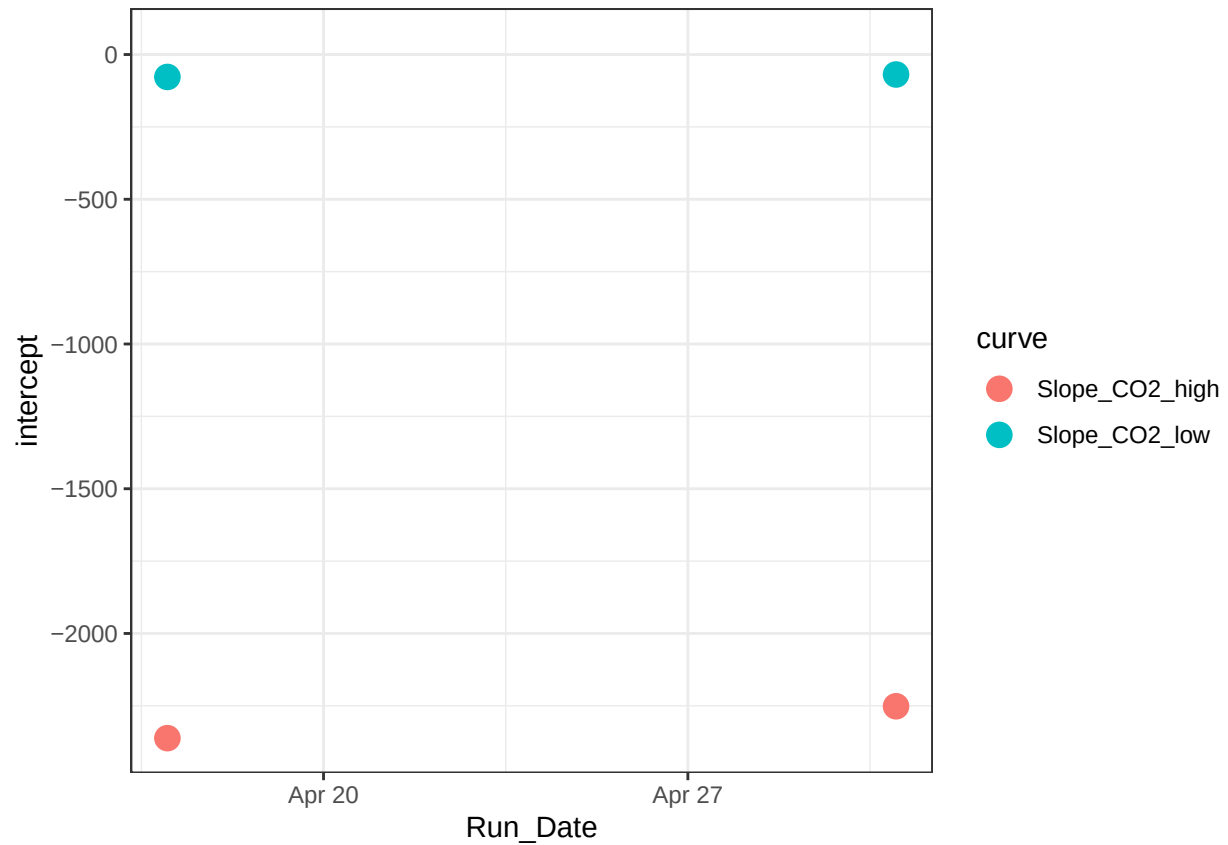
```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_C02 <- ggplot(unique_log %>% filter( grepl("C02",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_C02
```

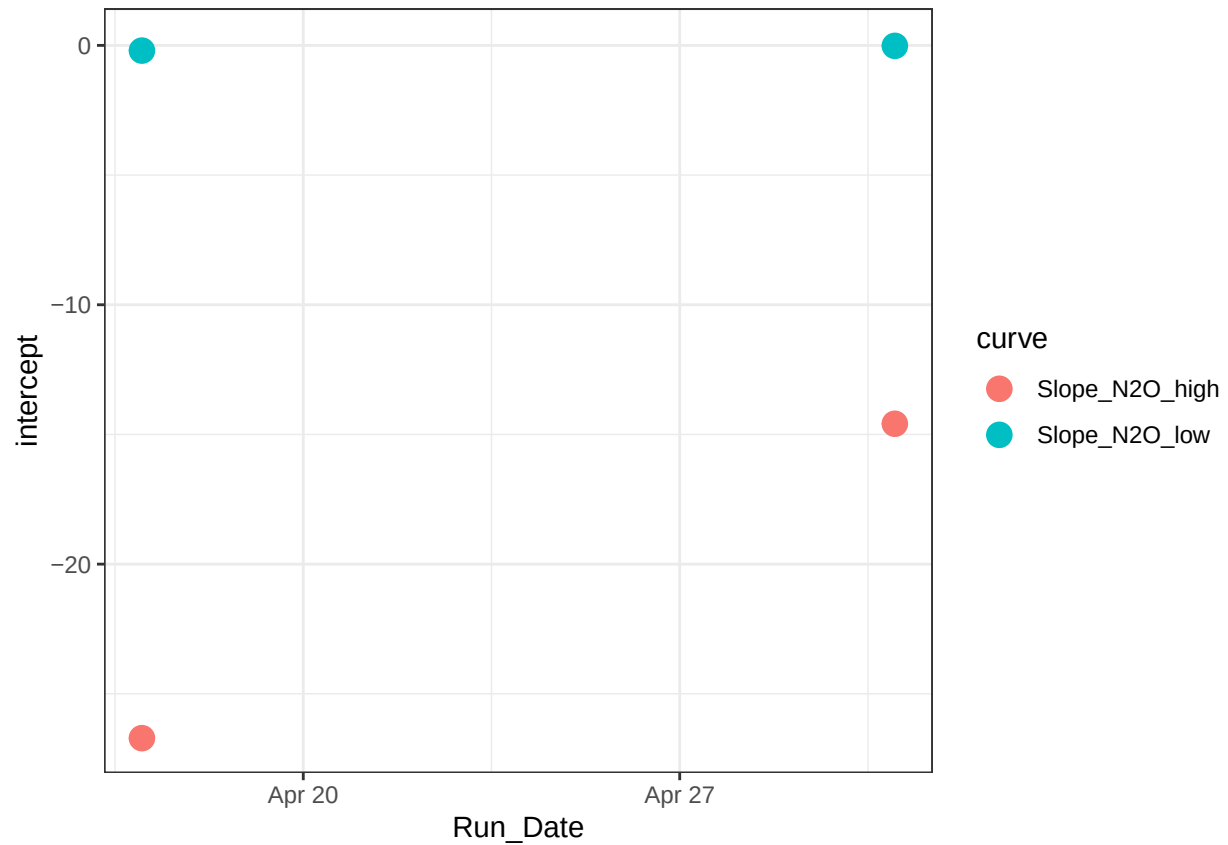
```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
int_chk_N20 <- ggplot(unique_log %>% filter( grepl("N20",curve,ignore.case = T)), aes(x = Run_Date, y = intercept)) +
  geom_point(size = 4) +
  #ylim(400, 800) +
  theme_bw()
```

```
int_chk_N20
```

```
## Warning: Removed 22 rows containing missing values or values outside the scale range
## ('geom_point()').
```



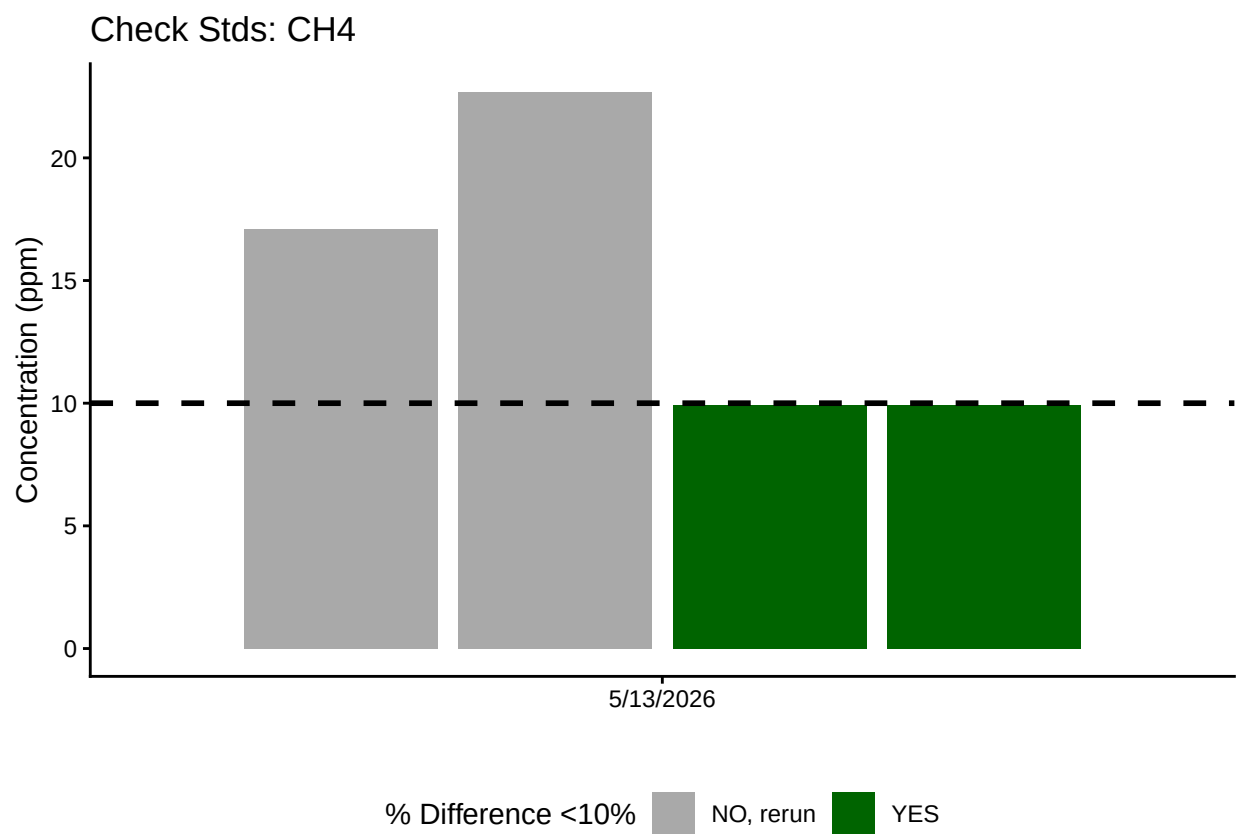
2.1 Now calculate the CH4 in ppm

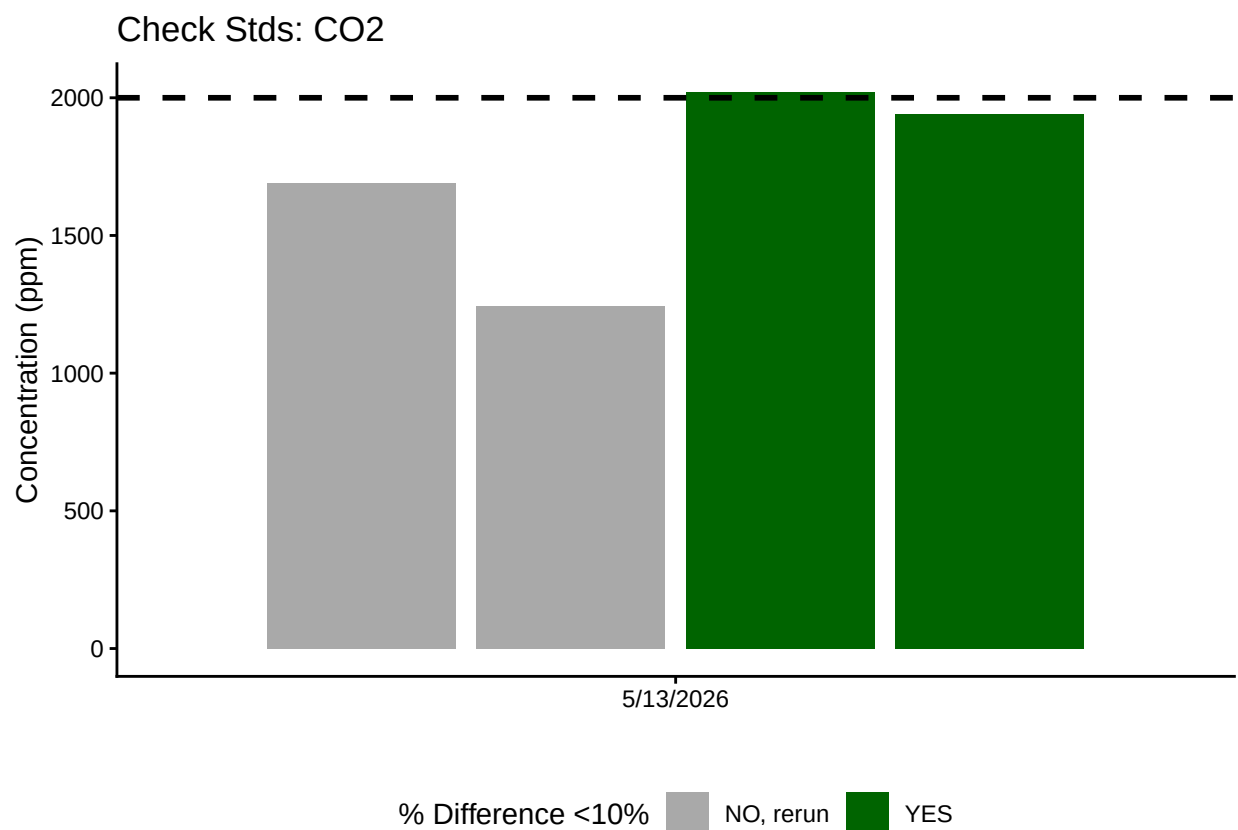
3 Aanlayze the Check Standards

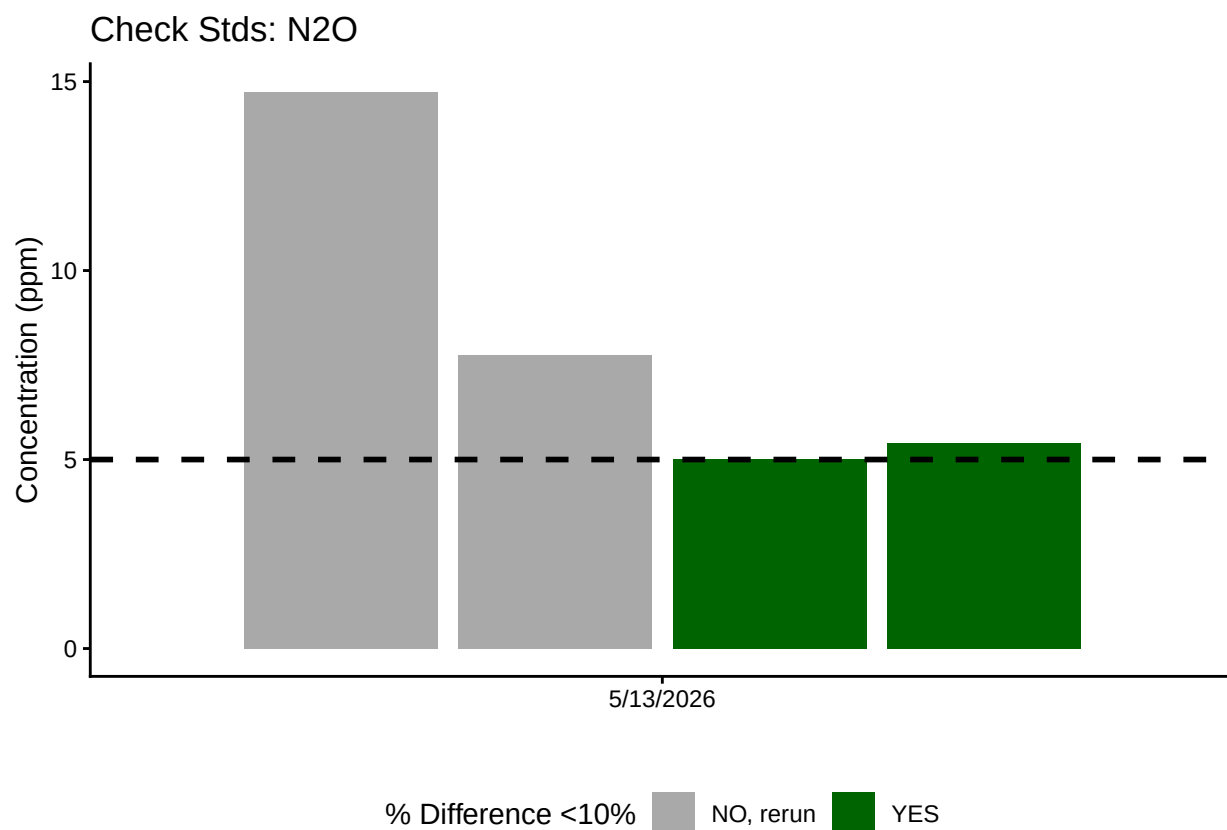
<60% of CH4 Check Standards are within range of expected concentration - REASSESS

<60% of CO2 Check Standards are within range of expected concentration - REASSESS

<60% of N2O Check Standards are within range of expected concentration - REASSESS







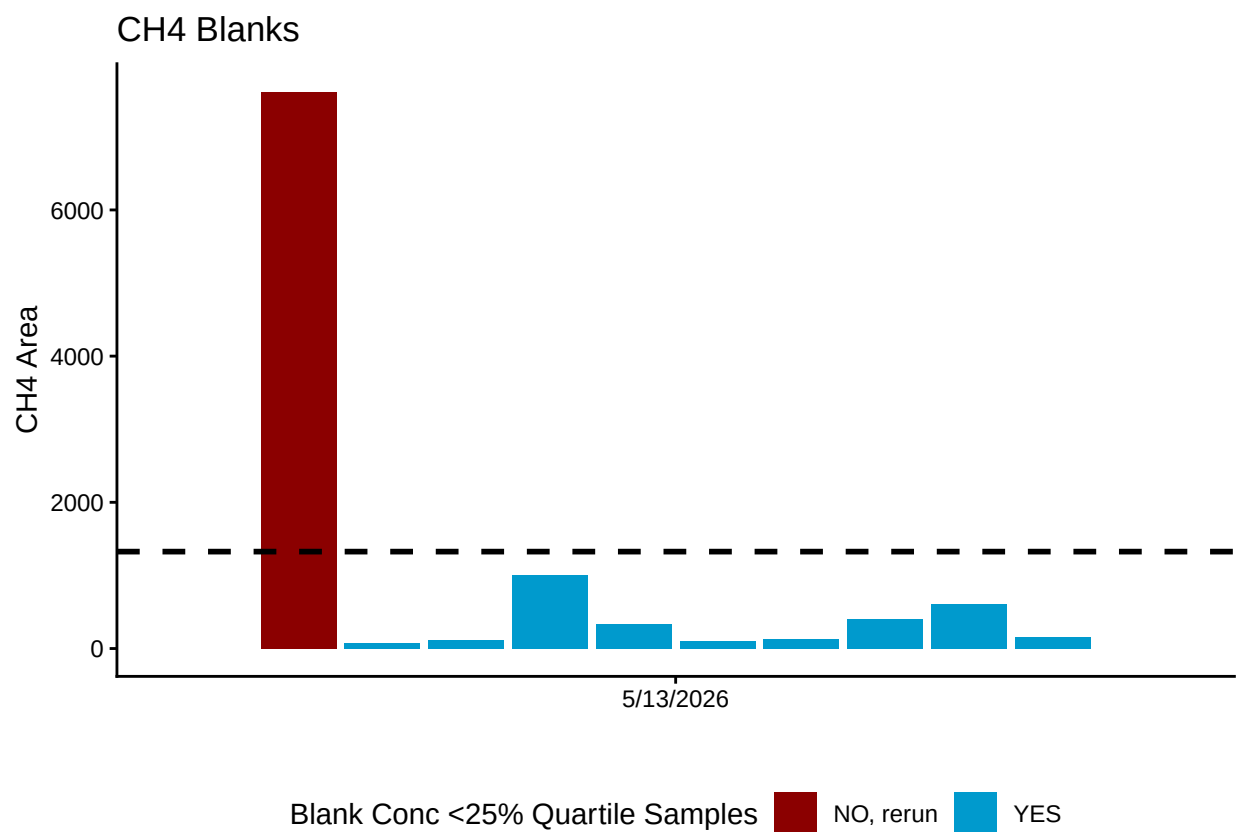
4 Analyze the Blanks

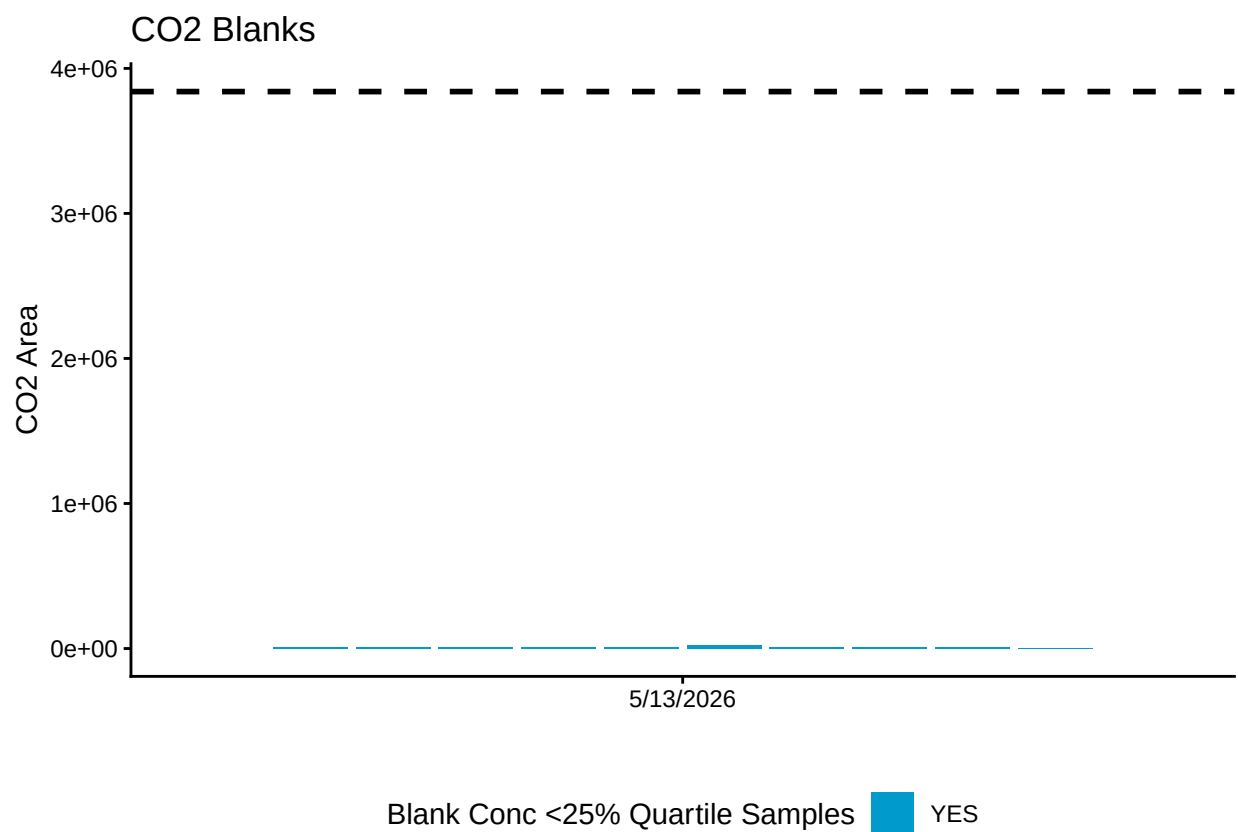
Assess Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of CO₂ Blanks are below the lower 25% quartile of samples - PROCEED

>60% of N₂O Blanks are below the lower 25% quartile of samples - PROCEED





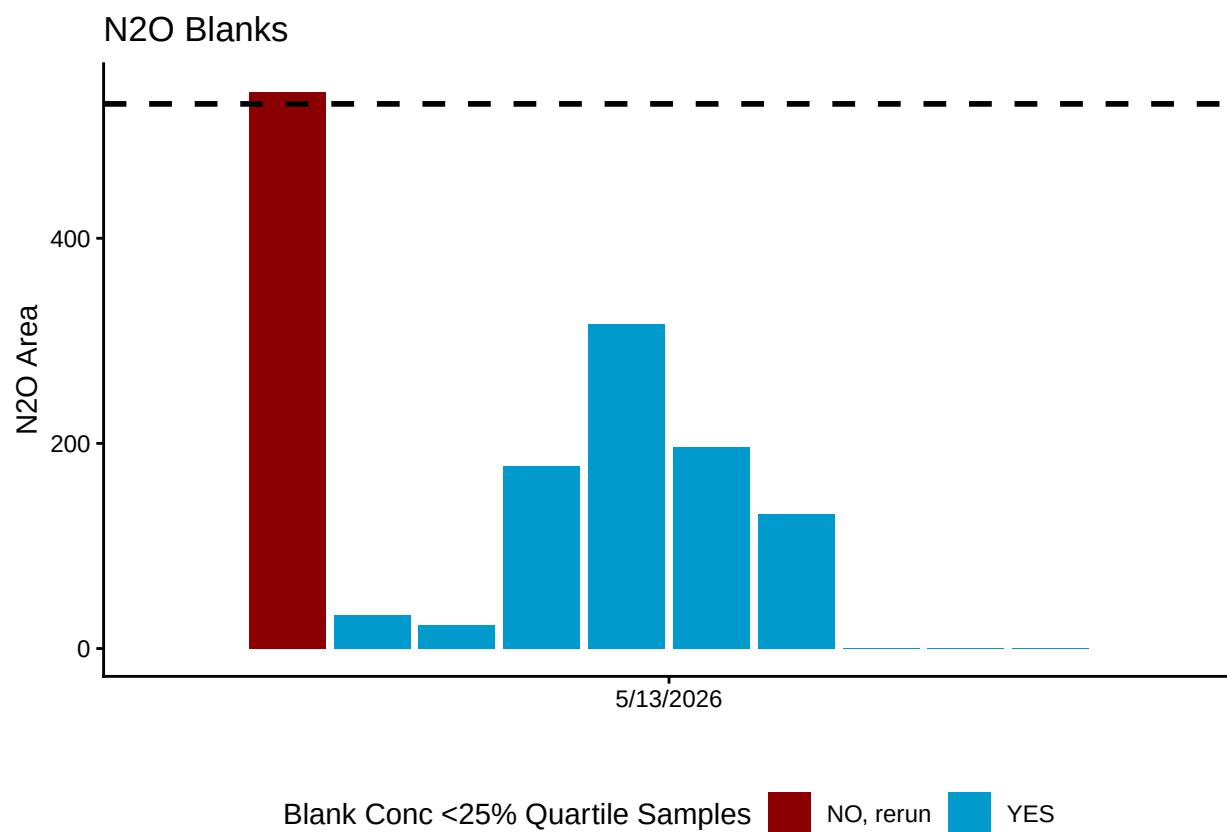
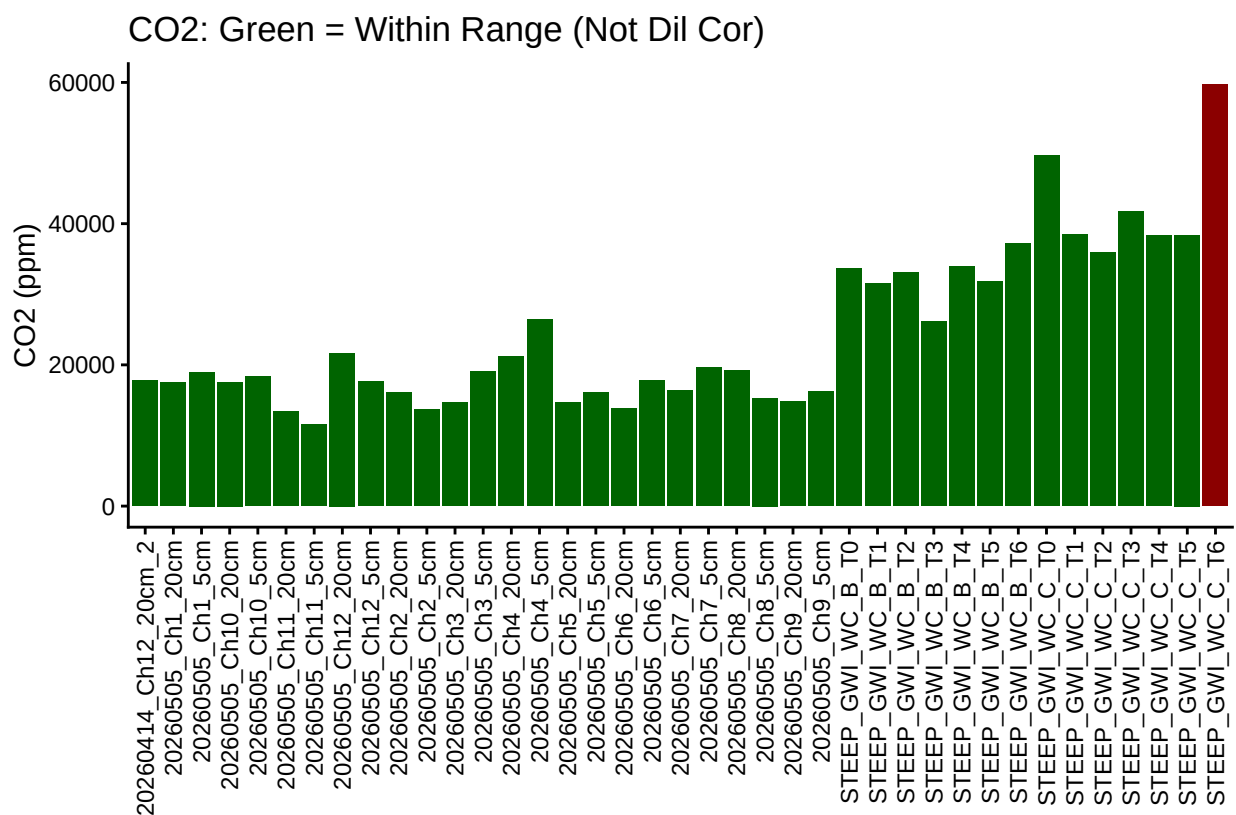
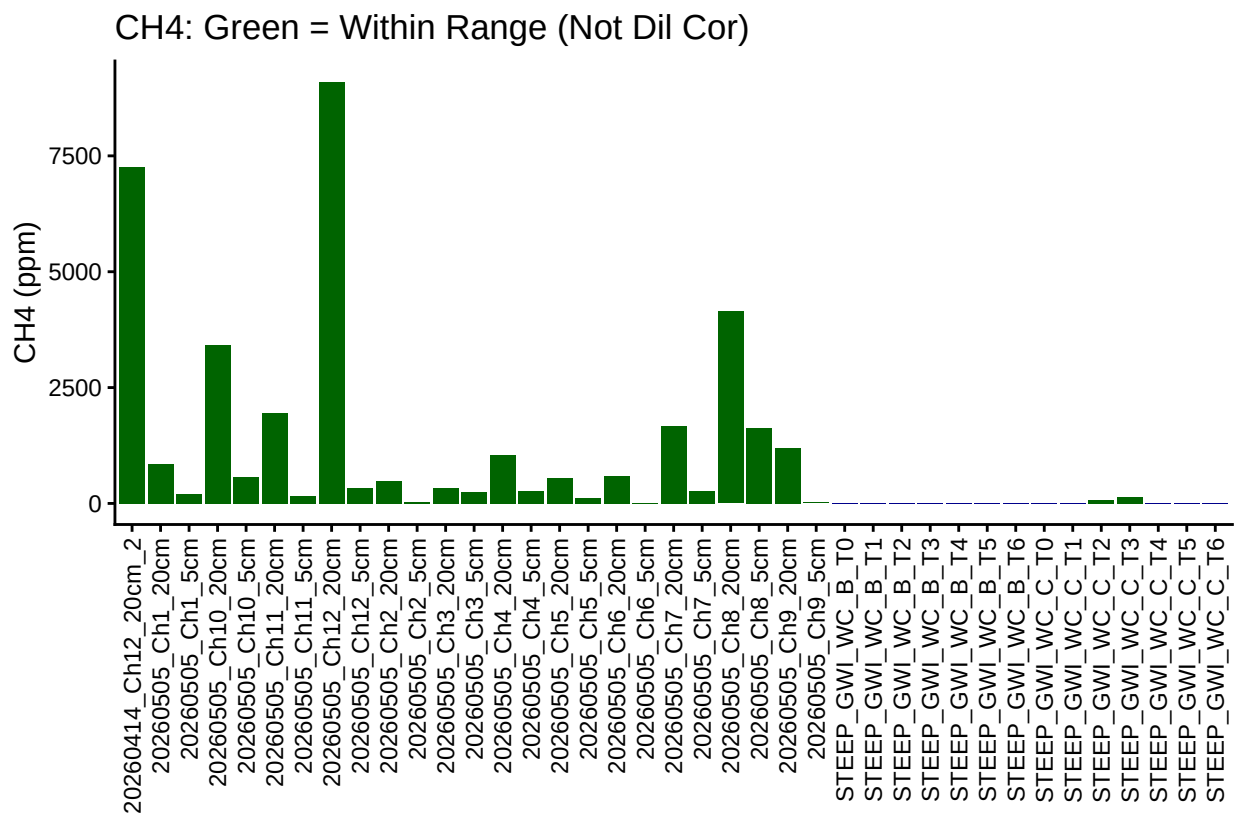
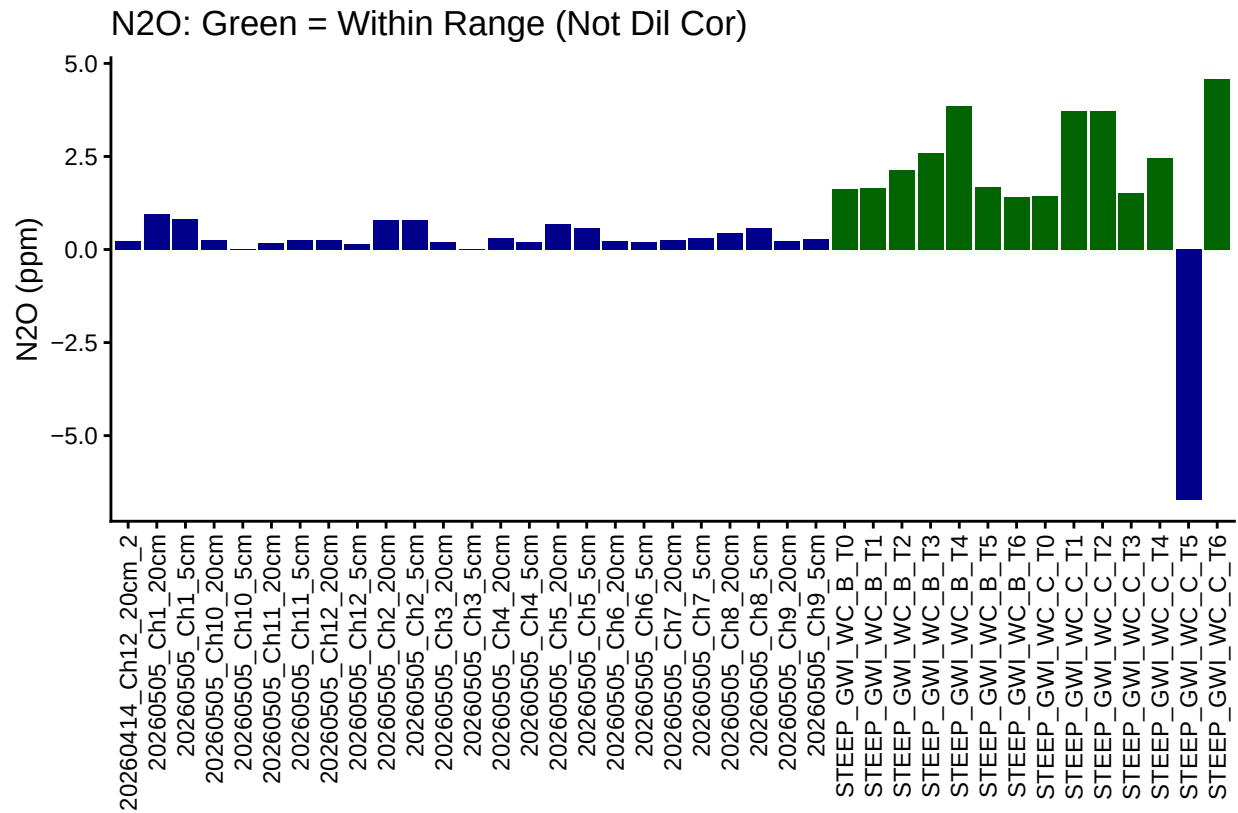


Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	1055.8
CO2	11286.6
N2O	142.0

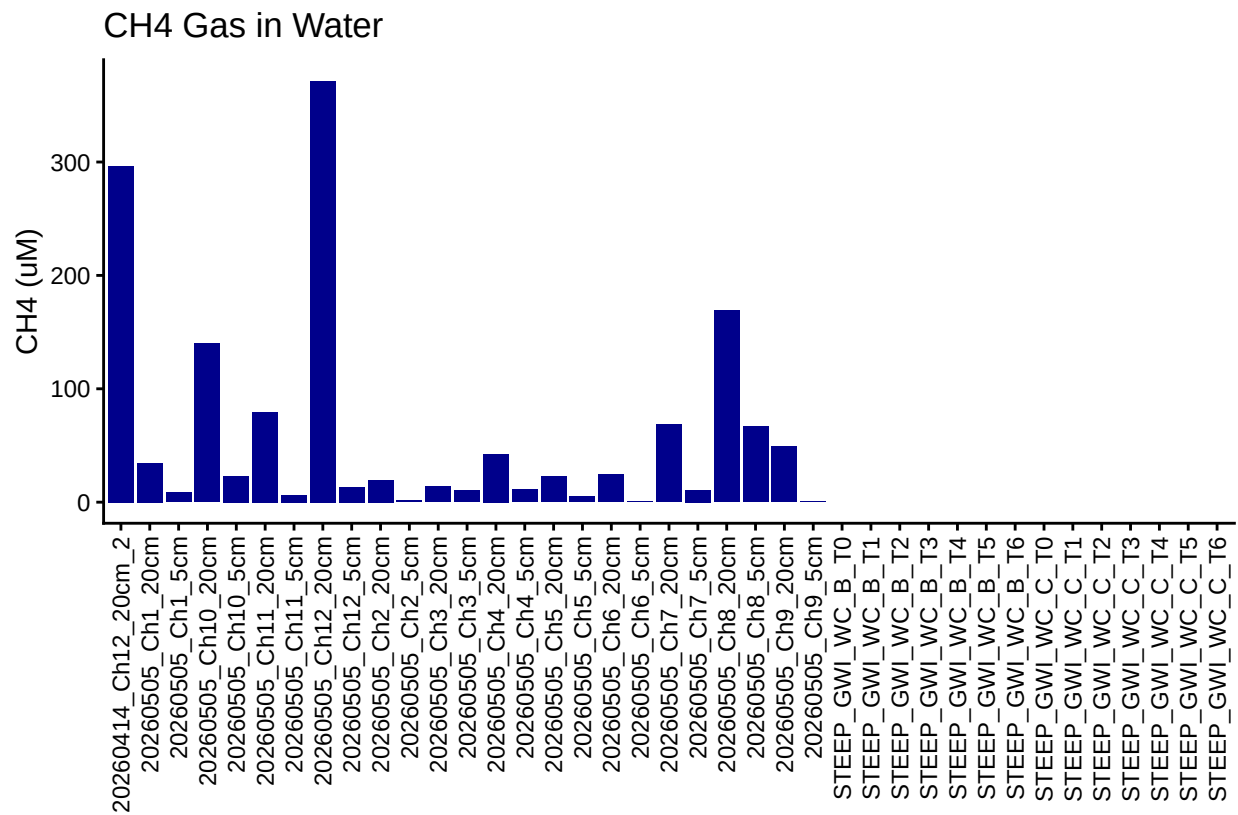
5 Dilution correct samples and Check Range



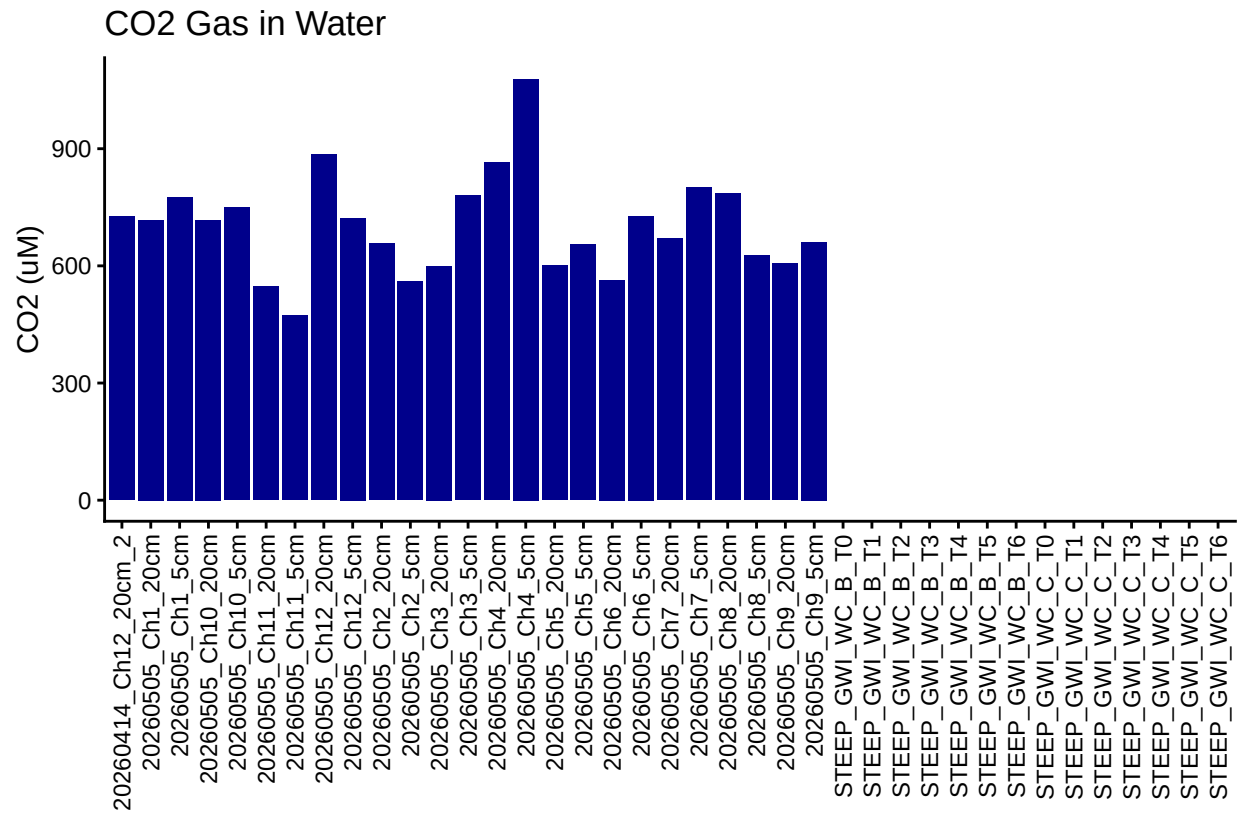


6 Calculate gas in water

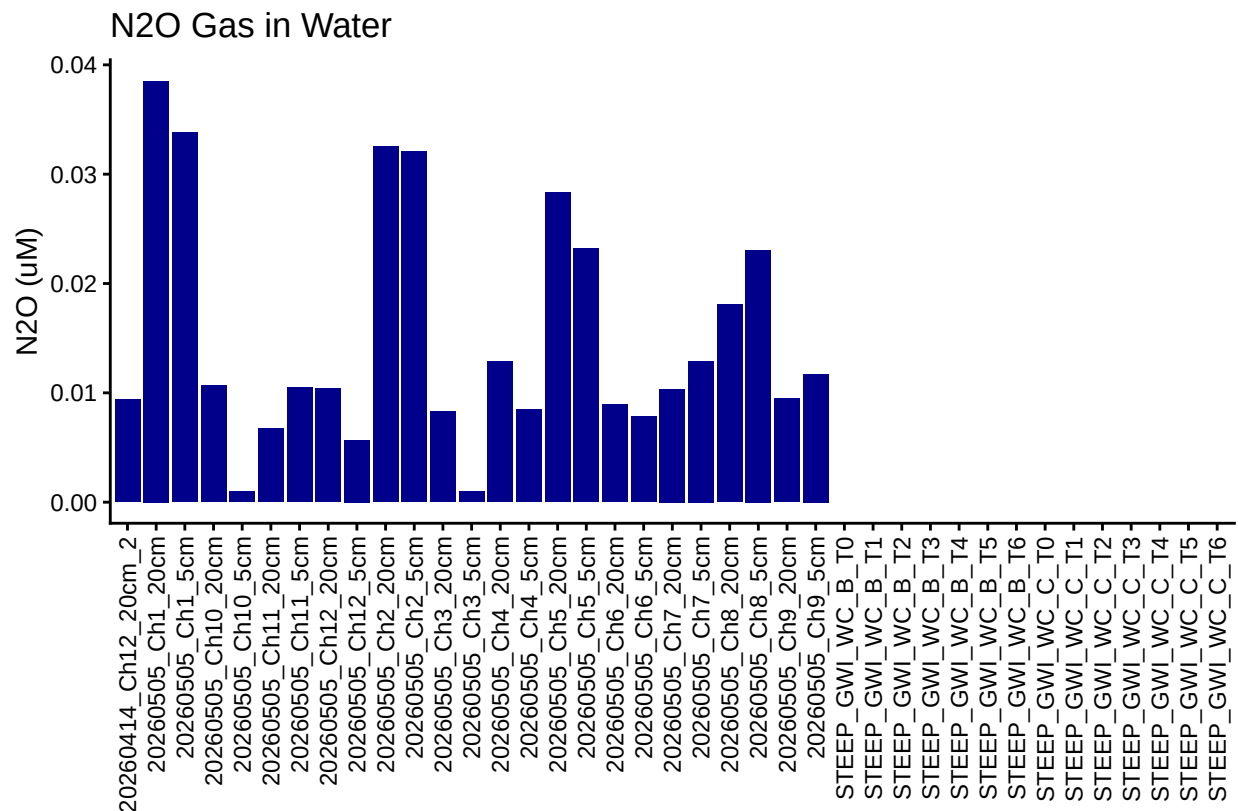
```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

```
## [1] "Ch1-40" "Ch1-80" "Ch2-40" "Ch2-80" "Ch3-40" "Ch3-80" "Ch4-40"
## [8] "Ch4-80" "Ch5-40" "Ch5-80" "Ch6-40" "Ch6-80" "Ch7-40" "Ch7-80"
## [15] "Ch8-40" "Ch8-80" "Ch9-40" "Ch9-80" "Ch10-40" "Ch10-80" "Ch11-40"
## [22] "Ch11-80" "Ch12-40" "Ch12-80"
```

7.1 Organize Data

```
## Run_Date N2O_Curve CO2_Curve CH4_Curve Sample_ID Type Conc_CO2
## 1 5/13/2026 High High Low STEEP_GWI_WC_C_T5 Sample unknown
## 2 5/13/2026 Low High High STEEP_GWI_WC_C_T3 Sample unknown
## 3 5/13/2026 Low High High STEEP_GWI_WC_C_T2 Sample unknown
## 4 5/13/2026 Low High High 20260505_Ch1_5cm Sample unknown
## 5 5/13/2026 Low High High 20260505_Ch1_20cm Sample unknown
## 6 5/13/2026 Low High High 20260505_Ch2_5cm Sample unknown
## Conc_CH4 Conc_N2O CO2_Area CH4_Area N2O_Area Notes Sample_Volume CH4_R2
## 1 unknown unknown 8496229 1320 14234 NA NA 0.9964451
## 2 unknown unknown 9192745 92429 3617 NA NA 0.9995580
```

##	3	unknown	unknown	7987736	49737	8984	NA	NA	0.9995580
##	4	unknown	unknown	4440043	133134	1945	NA	12	0.9995580
##	5	unknown	unknown	4140942	509564	2226	NA	12	0.9995580
##	6	unknown	unknown	3334986	32888	1844	NA	12	0.9995580
##		CH4_Slope	CH4_Intercept	CH4_Min	CH4_Max	CO2_R2	CO2_Slope	CO2_Intercept	
##	1	0.001637608	-0.1308363	2.5	10000	0.9927429	0.004784067	-2249.425	
##	2	0.001691193	-15.3776452	2.5	10000	0.9927429	0.004784067	-2249.425	
##	3	0.001691193	-15.3776452	2.5	10000	0.9927429	0.004784067	-2249.425	
##	4	0.001691193	-15.3776452	2.5	10000	0.9927429	0.004784067	-2249.425	
##	5	0.001691193	-15.3776452	2.5	10000	0.9927429	0.004784067	-2249.425	
##	6	0.001691193	-15.3776452	2.5	10000	0.9927429	0.004784067	-2249.425	
##		CO2_Min	CO2_Max	N2O_R2	N2O_Slope	N2O_Intercept	N2O_Min	N2O_Max	
##	1	500	50100	0.9940438	0.0007742149	-17.75508643	1.25	500	
##	2	500	50100	0.9969321	0.0004130590	0.02346392	1.25	500	
##	3	500	50100	0.9969321	0.0004130590	0.02346392	1.25	500	
##	4	500	50100	0.9969321	0.0004130590	0.02346392	1.25	500	
##	5	500	50100	0.9969321	0.0004130590	0.02346392	1.25	500	
##	6	500	50100	0.9969321	0.0004130590	0.02346392	1.25	500	
##		CH4_Conc_ppm	CO2_Conc_ppm	N2O_Conc_ppm	CH4_Flag	CO2_Flag	N2O_Flag		
##	1	2.030807	38397.10	-6.7349114	BDL	Within Range	BDL		
##	2	140.937620	41729.28	1.5174982	Within Range	Within Range	Within Range		
##	3	68.737214	35964.44	3.7343858	Within Range	Within Range	Within Range		
##	4	209.777625	18992.04	0.8268636	Within Range	Within Range	BDL		
##	5	846.393354	17561.12	0.9429332	Within Range	Within Range	BDL		
##	6	40.242306	13705.37	0.7851447	Within Range	Within Range	BDL		
##		CH4_Conc_ppm_dilcorr	CO2_Conc_ppm_dilcorr	N2O_Conc_ppm_dilcorr	CH4_Conc_uM				
##	1		NA	NA	NA	NA			
##	2		NA	NA	NA	NA			
##	3		NA	NA	NA	NA			
##	4		209.77763	18992.04	0.8268636	8.574185			
##	5		846.39335	17561.12	0.9429332	34.594409			
##	6		40.24231	13705.37	0.7851447	1.644813			
##		CO2_Conc_uM	N2O_Conc_uM						
##	1		NA	NA					
##	2		NA	NA					
##	3		NA	NA					
##	4		776.2565	0.03379618					
##	5		717.7709	0.03854026					
##	6		560.1760	0.03209101					

##		Site	Project	Year	Month	Sample_Date	Community	Nickname	Plot	Depth_cm
##	1	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch1	5
##	2	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch1	20
##	3	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch2	5
##	4	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch2	20
##	5	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch3	5
##	6	GCRew	GENX	2026	May	2026-05-05	SCAM, SPPA, DISP	GENX	Ch3	20
##		CO2_Treatment	Temp_Treatment	CO2_Conc_ppm	CO2_Conc_uM	CO2_Conc_Flag				
##	1	Ambient		0.00	18992.04	776.2565	Within Range			
##	2	Ambient		0.00	17561.12	717.7709	Within Range			
##	3	Ambient		0.00	13705.37	560.1760	Within Range			
##	4	Ambient		0.00	16069.14	656.7897	Within Range			
##	5	Ambient		0.75	19072.20	779.5330	Within Range			
##	6	Ambient		0.75	14674.51	599.7873	Within Range			

	N2O_Conc_ppm	N2O_Conc_uM	N2O_Conc_Flag	CH4_Conc_ppm	CH4_Conc_uM
## 1	0.82686363	0.0337961758	BDL	209.77763	8.574185
## 2	0.94293320	0.0385402564	BDL	846.39335	34.594409
## 3	0.78514467	0.0320910081	BDL	40.24231	1.644813
## 4	0.79588420	0.0325299622	BDL	482.41314	19.717543
## 5	0.02346392	0.0009590345	BDL	245.66136	10.040851
## 6	0.20273151	0.0082861910	BDL	334.96141	13.690788

##	CH4_Conc_Flag
## 1	Within Range
## 2	Within Range
## 3	Within Range
## 4	Within Range
## 5	Within Range
## 6	Within Range

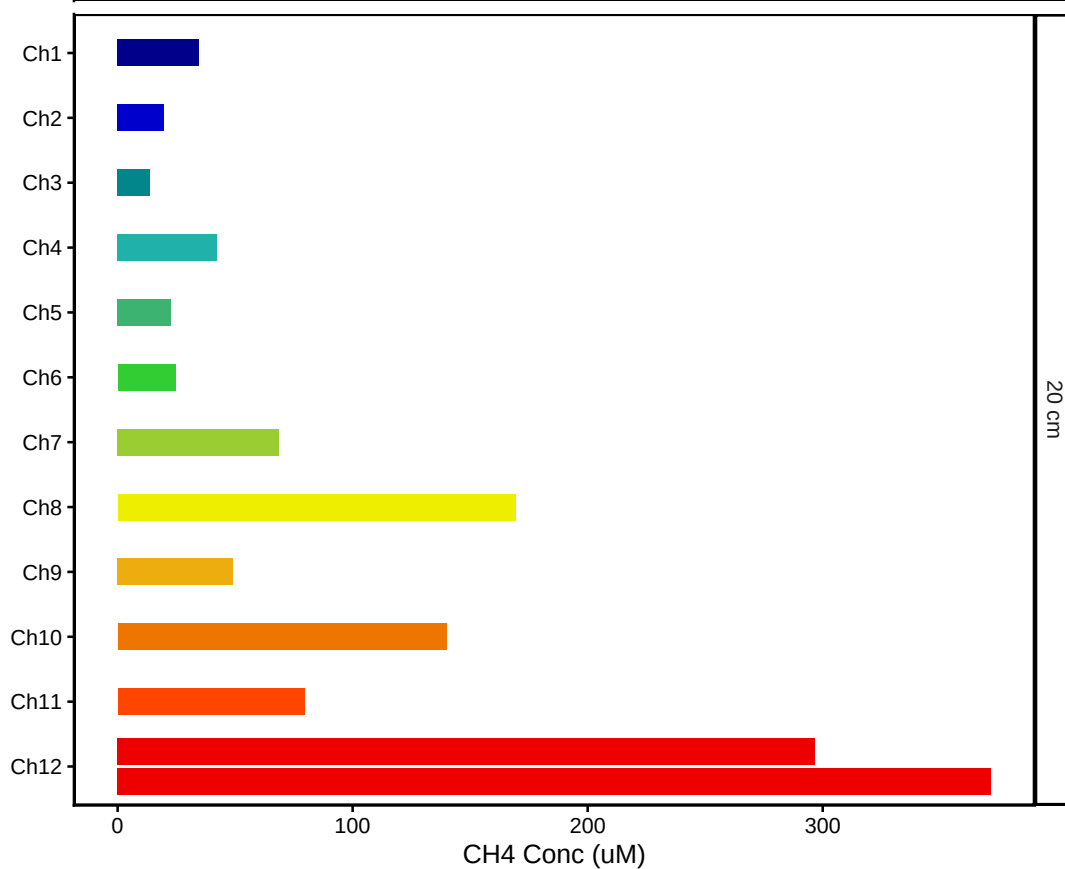
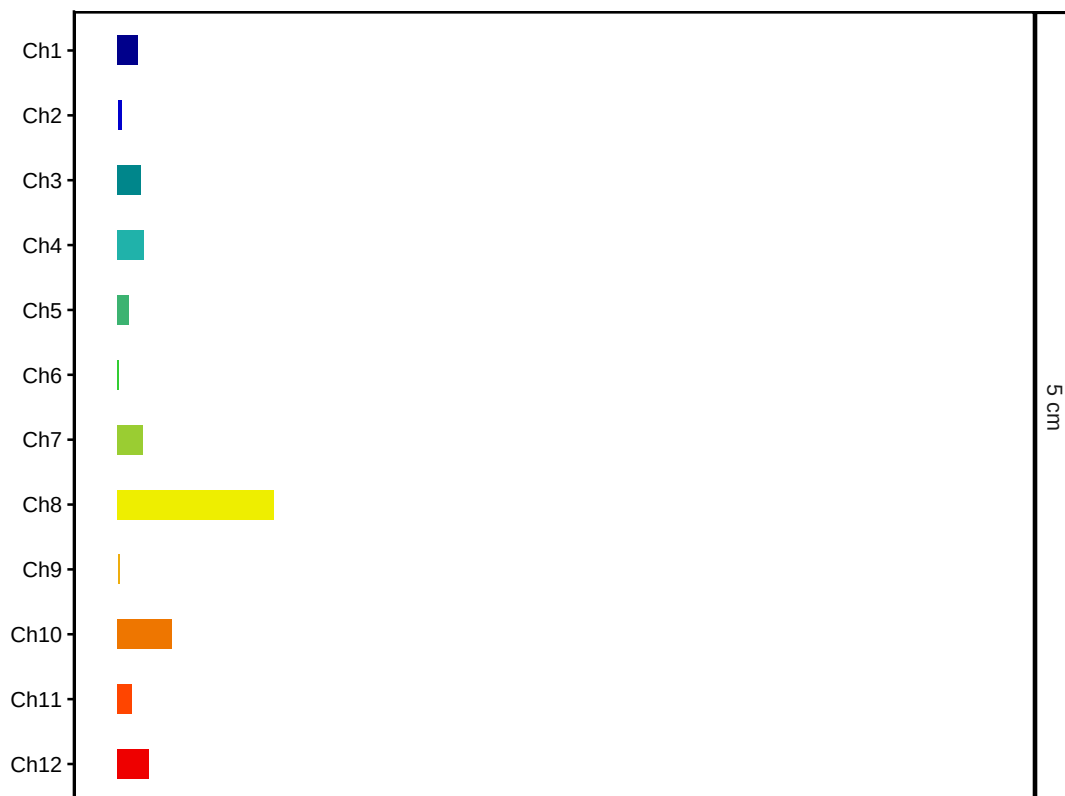
##	QAQC_F
## 1	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range
## 2	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range
## 3	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range
## 4	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range
## 5	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range
## 6	; CH4 Check Standards out Of Range; CO2 Check Standards out Of Range; N2O Check Standards out Of Range

##	Analysis_rundate	Run_notes
## 1	5/13/2026	
## 2	5/13/2026	
## 3	5/13/2026	
## 4	5/13/2026	
## 5	5/13/2026	
## 6	5/13/2026	

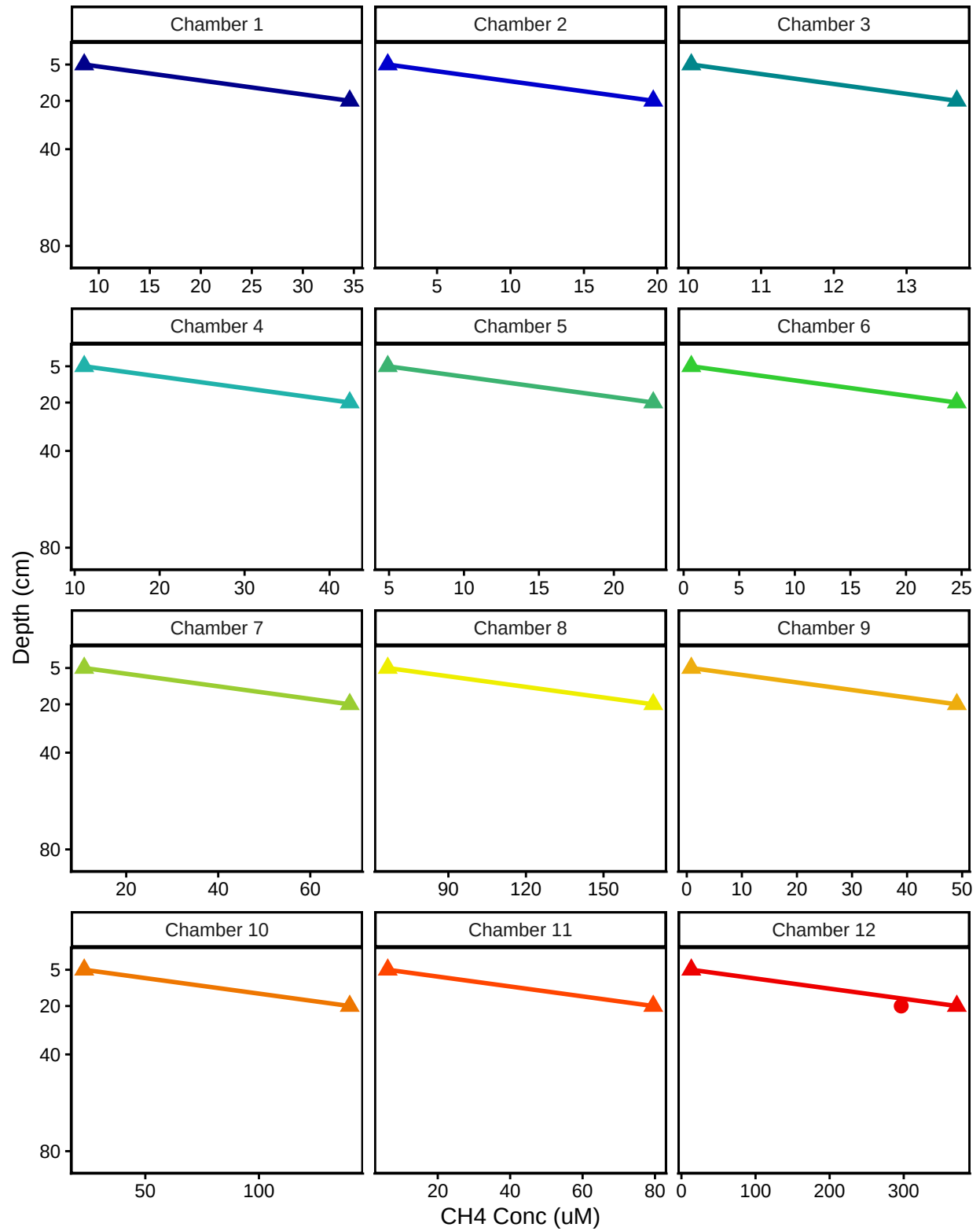
7.2 Visualize Data CH4

Visualize Data

GENX: Porewater CH4



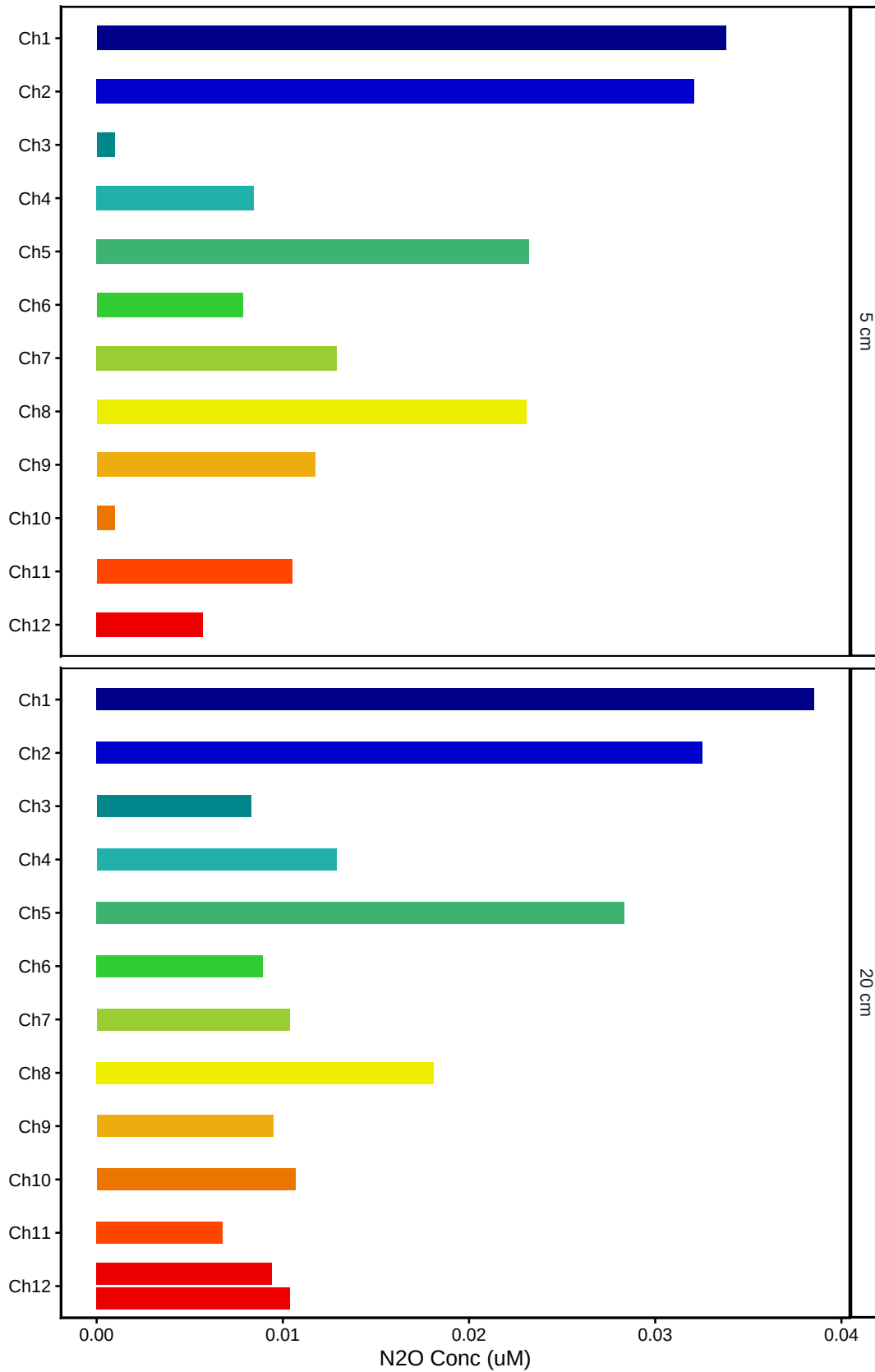
GENX: Porewater CH4 by Chamber



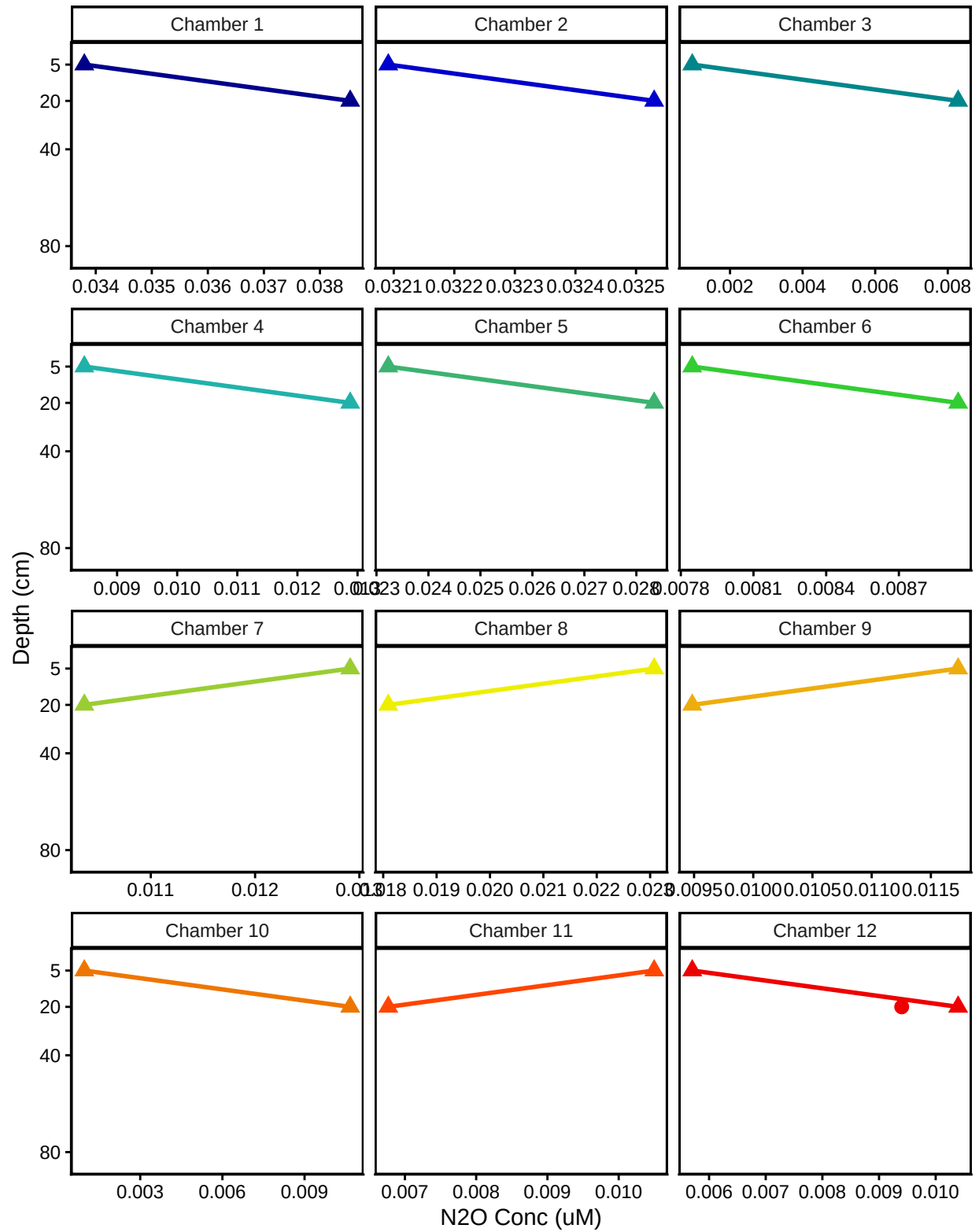
##Visualize Data N2O

```
## Visualize Data
```

GENX: Porewater N₂O



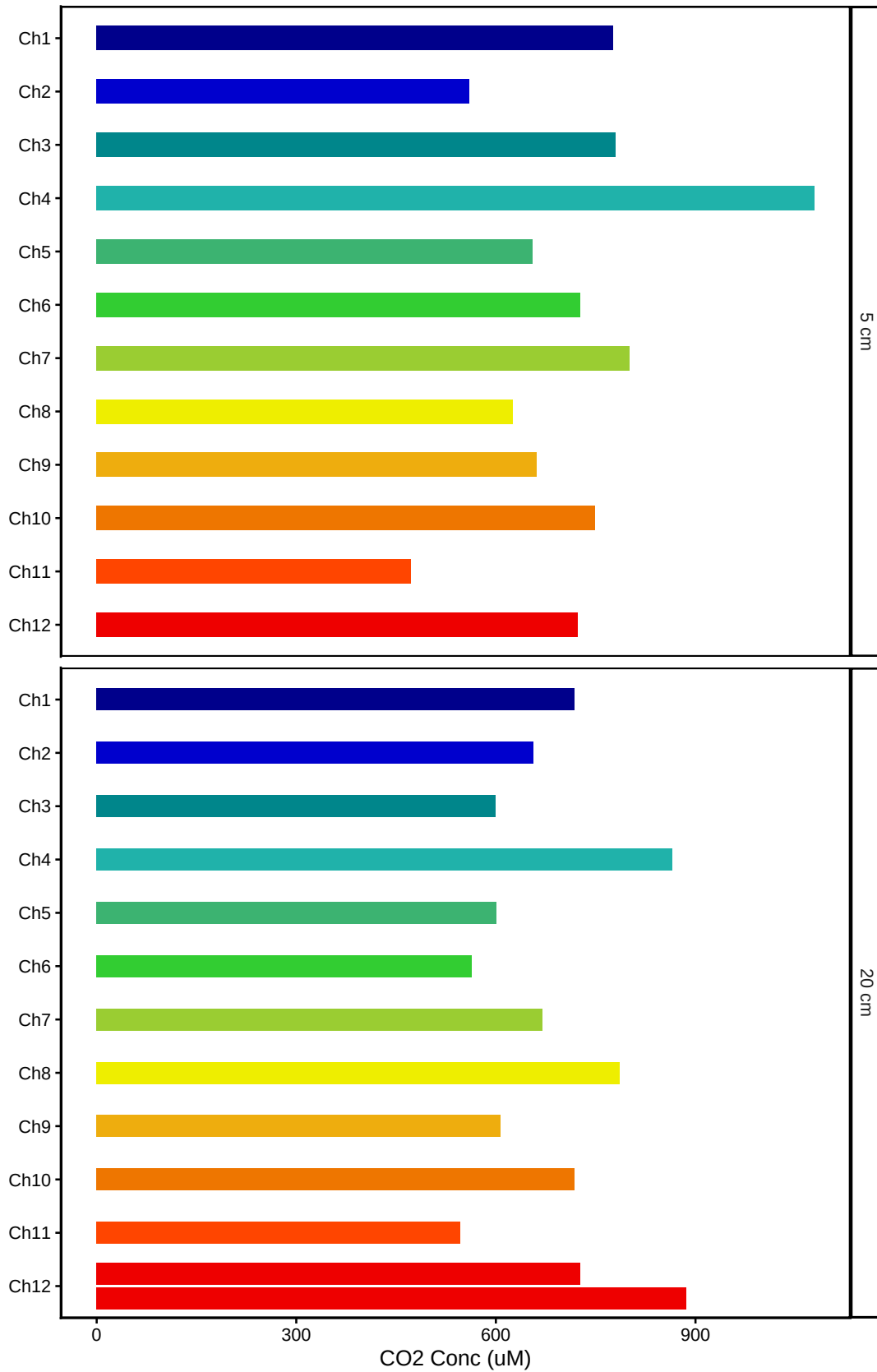
GENX: Porewater N2O by Chamber



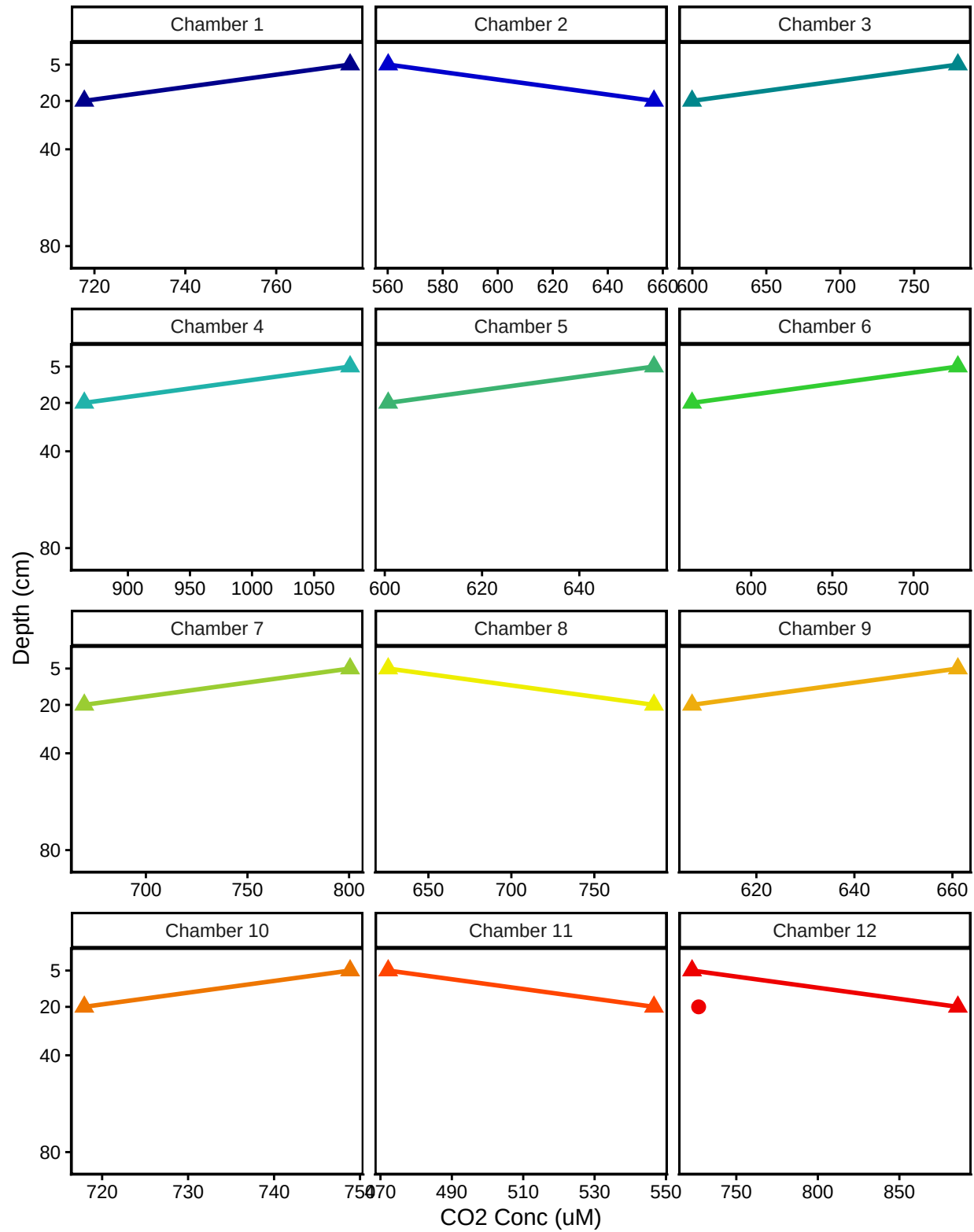
##Visualize Data CO2

```
## Visualize Data
```

GENX: Porewater CO2



GENX: Porewater CO2 by Chamber



##Export Data

#end